Augmenting Humans

From Physiology to Shared Experiences

Designing feedback loops in the age of AI

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Human-Centered Information Systems · TU Clausthal
kaikunze.de



TU Clausthal GEIST

Academic Mindtrek 2026 · Tampere

Saraiji et al. MetaArms: body remapping using feet-controlled artificial arms. UIST 2018

AI Does More of the Work

What happens to the person?



Thinking

319 knowledge workers: more trust in GenAI, less critical thinking.

Motivation

N = 3,562: better work with GenAI, less motivation once back on their own.

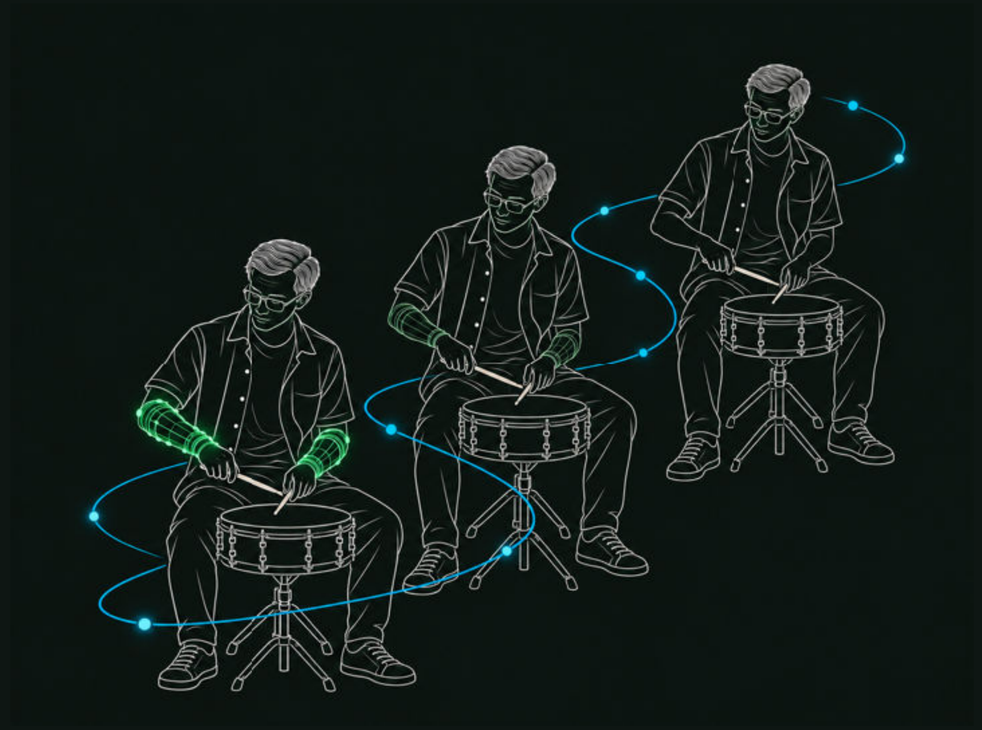
Ownership

n = 30 + 96: people did not feel they owned AI-written text, yet signed it.

Lee et al., CHI 2025 · Wu et al., Scientific Reports 15:15105, 2025 · Draxler, …, Schmidt, Buschek, Welsch, ACM TOCHI 2023

We Design the Loops/Helixes/Coupling/Resonance …

AI takes on the task; the feedback is ours to design



Agency

The system senses how I engage. It still feels like me doing it.

Skill

Which skills do I need? Which should I develop myself?

Experience

Meaningful moments with others, and ways to share them.

Engelbart, Augmenting Human Intellect, SRI 1962 · Ryan & Deci, American Psychologist 55(1), 2000

Augmenting Humans

Overview



Personal background

Wearable computing since 2000



Activity recognition

Computers learning what we do



System development

Glasses and sensors people wear all day



Augmented

Agency · Skill · Experience



My Background

Professor (2026)

Professor (2018)

Research Assistant Professor (2012)

Visiting Researcher at MIT Media Lab

PhD in Ubiquitous/Wearable Computing (2011)

Collaborations with PARC, SUN, Deutsche Börse ...

Human-Centered Information Systems, TU Clausthal

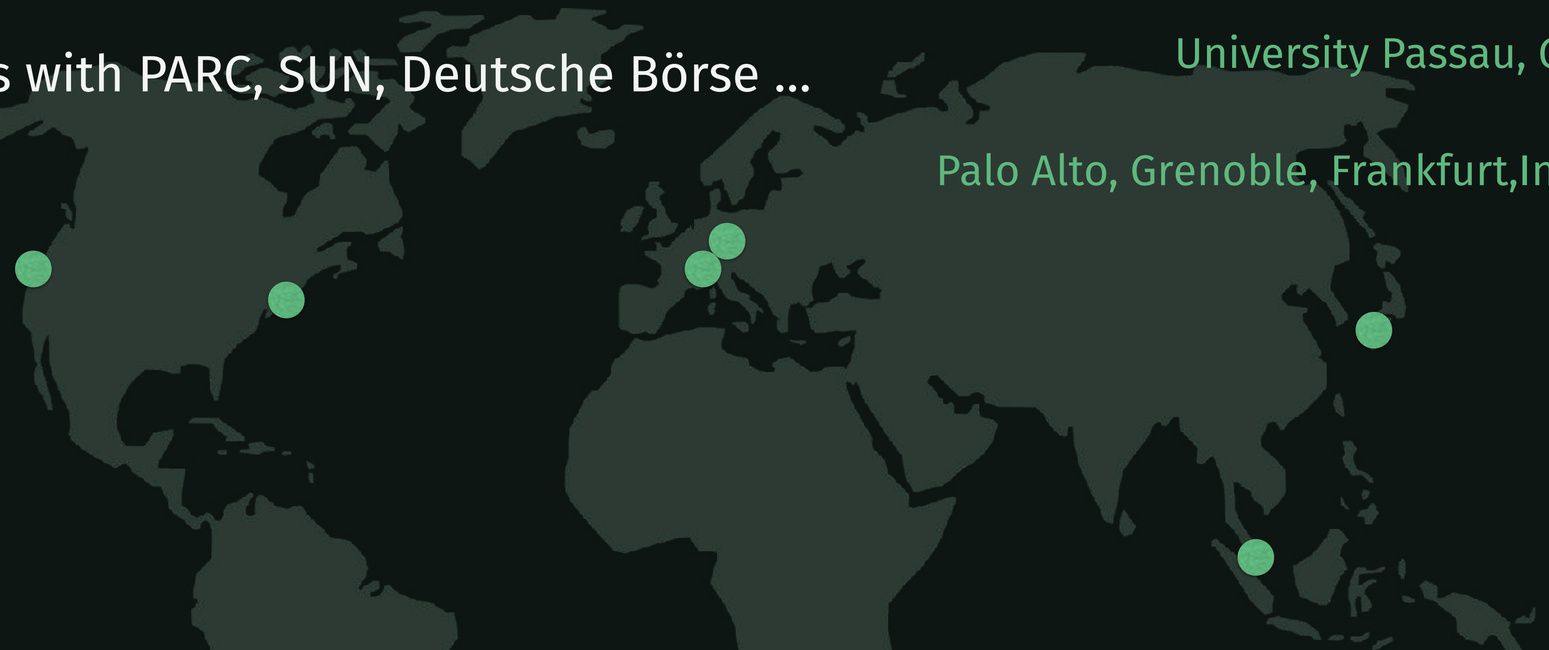
Graduate School of Media Design, Keio University

Osaka Prefecture University

MIT, Cambridge, USA

University Passau, Germany

Palo Alto, Grenoble, Frankfurt,Innsbruck





Our group at TU Clausthal

Still with us
in Yokohama (Keio)



Mingyang Xu



Danyang Peng



Kanyu Chen

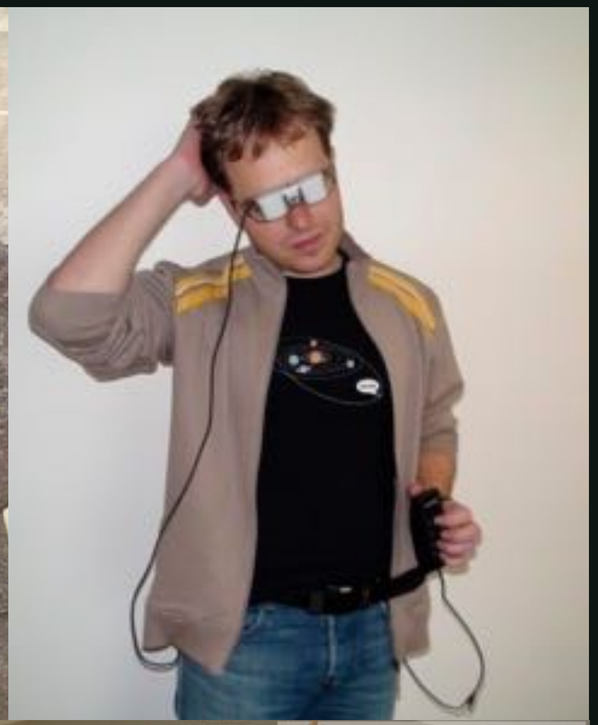
Activity Recognition



2000: Worn All Day, Already a Loop

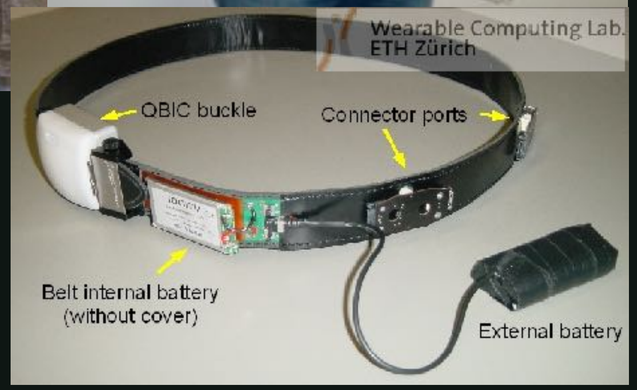






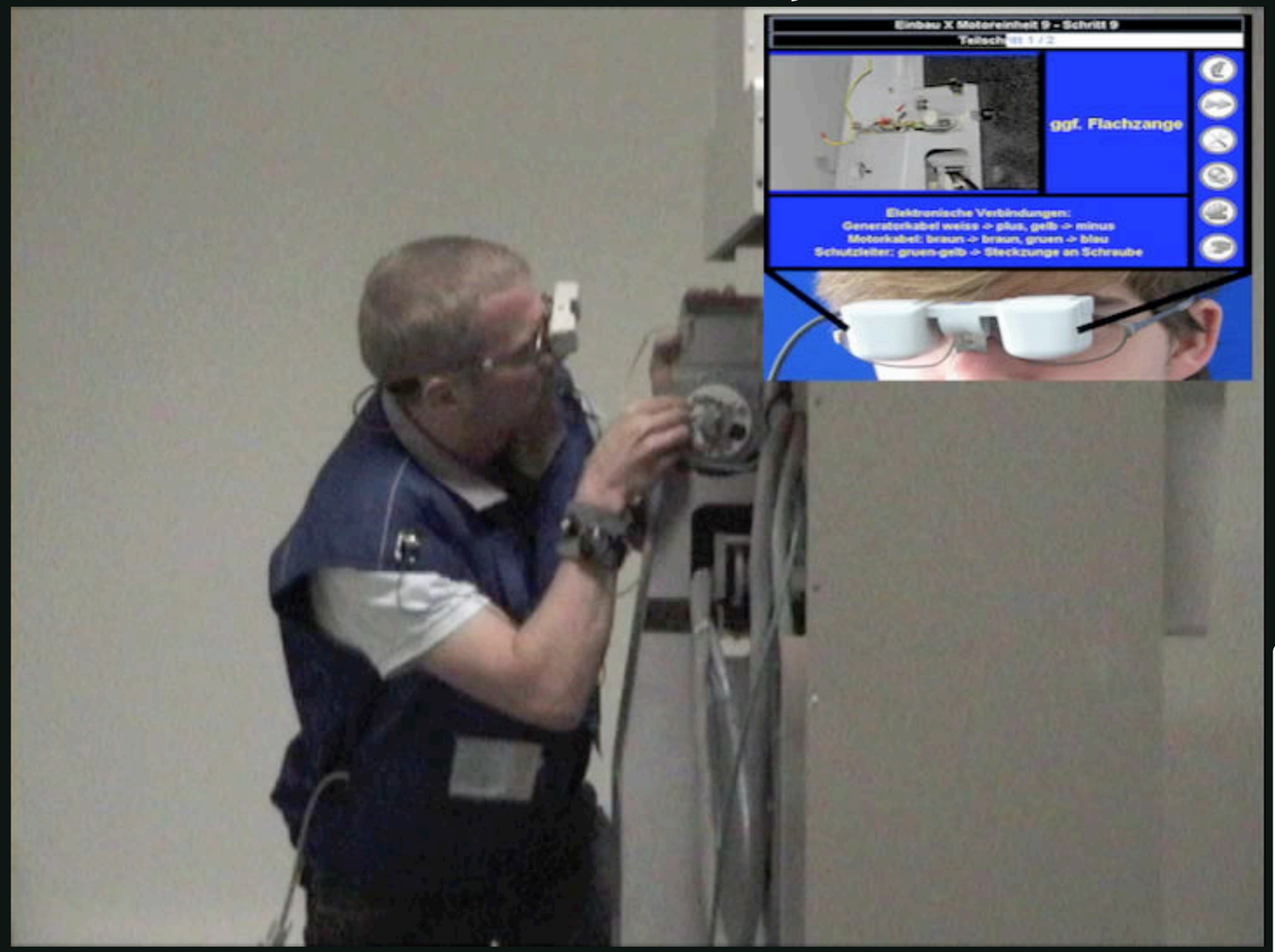




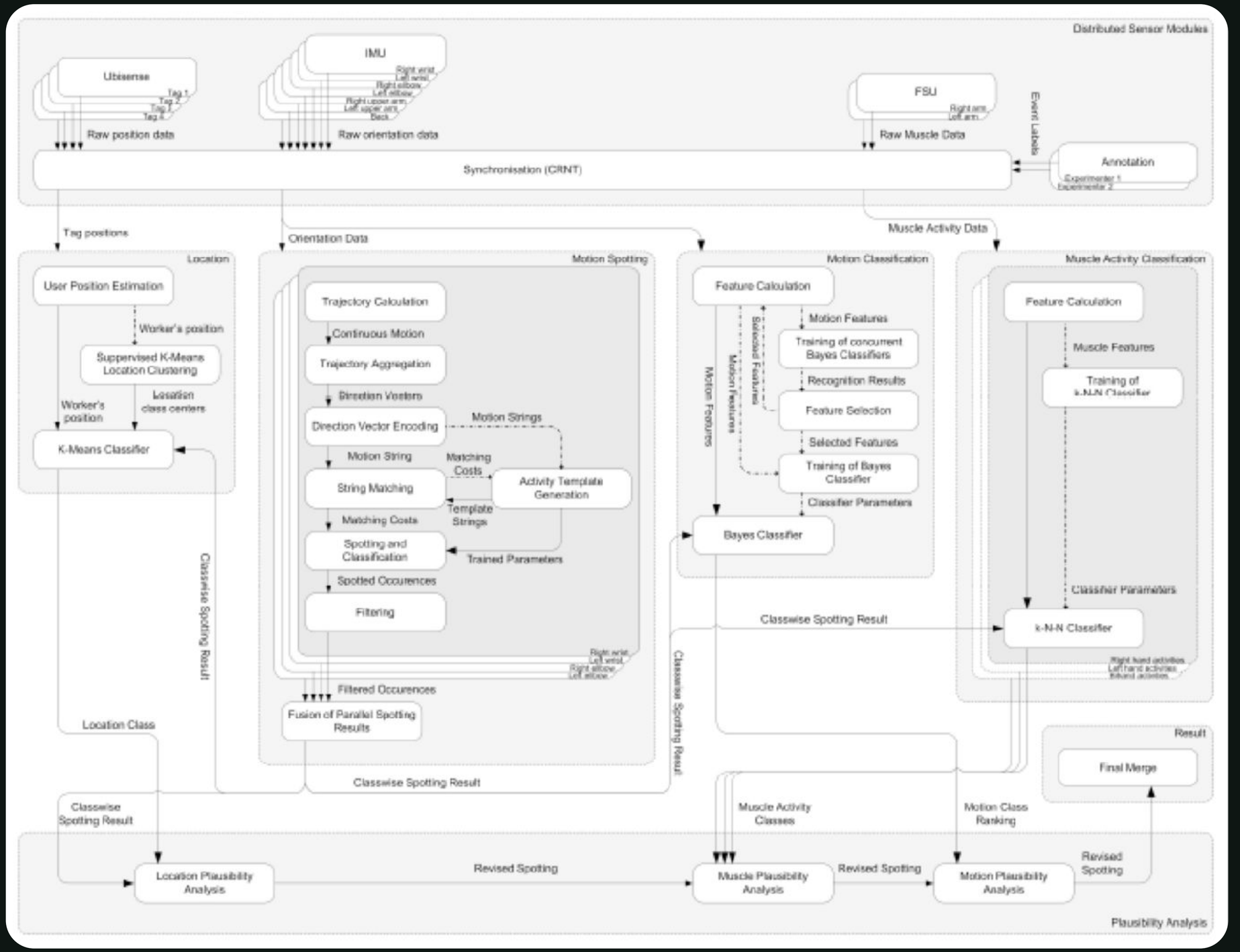


2007: The System Tells You the Next Step

Maintenance Scenario, Collaboration with University Bremen and Zeiss.



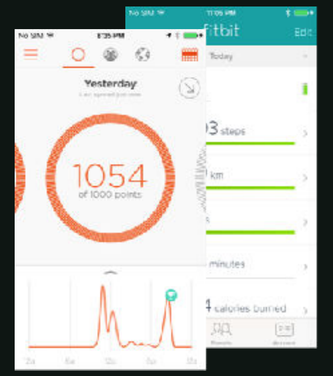
Kunze, Kai, et al. "Does context matter?-a quantitative evaluation in a real world maintenance scenario." Pervasive Computing: 7th International Conference, 2009.


Distributed Sensor Modules
Ubisense
Tag 1
Tag 2
Tag 3
Tag 4
Raw position data
IMU
Right wrist
Left wrist
Right elbow
Left elbow
Right upper arm
Left upper arm
Back
Raw orientation data
FSU
Right arm
Left arm
Raw Muscle Data
Event Labels
Annotation
Experimenter 1
Experimenter 2
Synchronisation (CRNT)
Tag positions
Orientation Data
Muscle Activity Data
Location
User Position Estimation
Worker's position
Supervised K-Means Location Clustering
Worker's position
Location class centers
K-Means Classifier
Classwise Spotting Result
Location Class
Motion Spotting
Trajectory Calculation
Continuous Motion
Trajectory Aggregation
Direction Vectors
Direction Vector Encoding
Motion Strings
Motion String
Matching Costs
String Matching
Activity Template Generation
Template Strings
Matching Costs
Spotting and Classification
Trained Parameters
Spotted Occurences
Filtering
Filtered Occurences
Fusion of Parallel Spotting Results
Classwise Spotting Result
Motion Classification
Feature Calculation
Motion Features
Training of concurrent Bayes Classifiers
Recognition Results
Feature Selection
Selected Features
Training of Bayes Classifier
Classifier Parameters
Bayes Classifier
Classwise Spotting Result
Muscle Activity Classification
Feature Calculation
Muscle Features
Training of k-NN Classifier
Classifier Parameters
k-N-N Classifier
Classwise Spotting Result
Result
Final Merge
Classwise Spotting Result
Location Plausibility Analysis
Revised Spotting
Muscle Activity Classes
Muscle Plausibility Analysis
Revised Spotting
Motion Class Ranking
Motion Plausibility Analysis
Revised Spotting
Plausibility Analysis

Where You Wear It Changes What You See











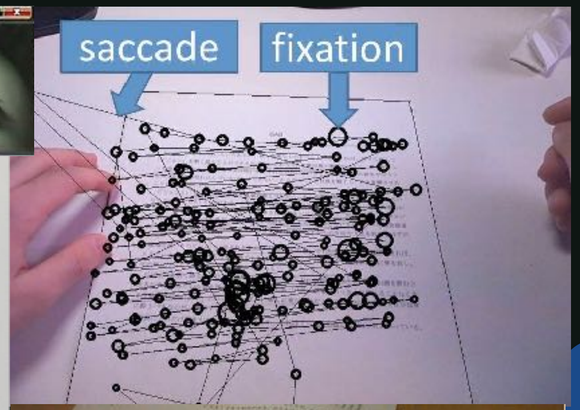




Kunze Kai. Compensating for On-Body Placement Effects in Activity Recognition, 2011.

From Eye Movements Alone: What You Are Reading





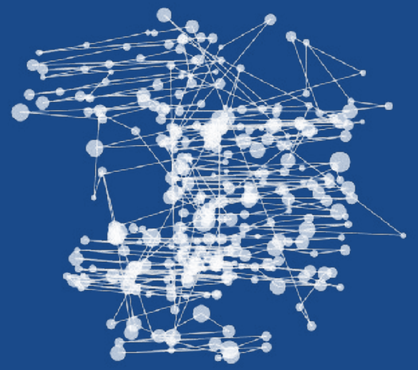


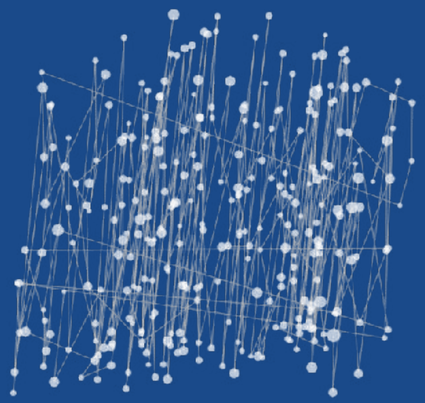


Kai Kunze, Yuzuko Utsumi, Yuki Shiga, Koichi Kise, and Andreas Bulling. 2013. I know what you are reading: recognition of document types using mobile eye tracking. ISWC '13

Where to start?

Ubiquitous learning activity: reading



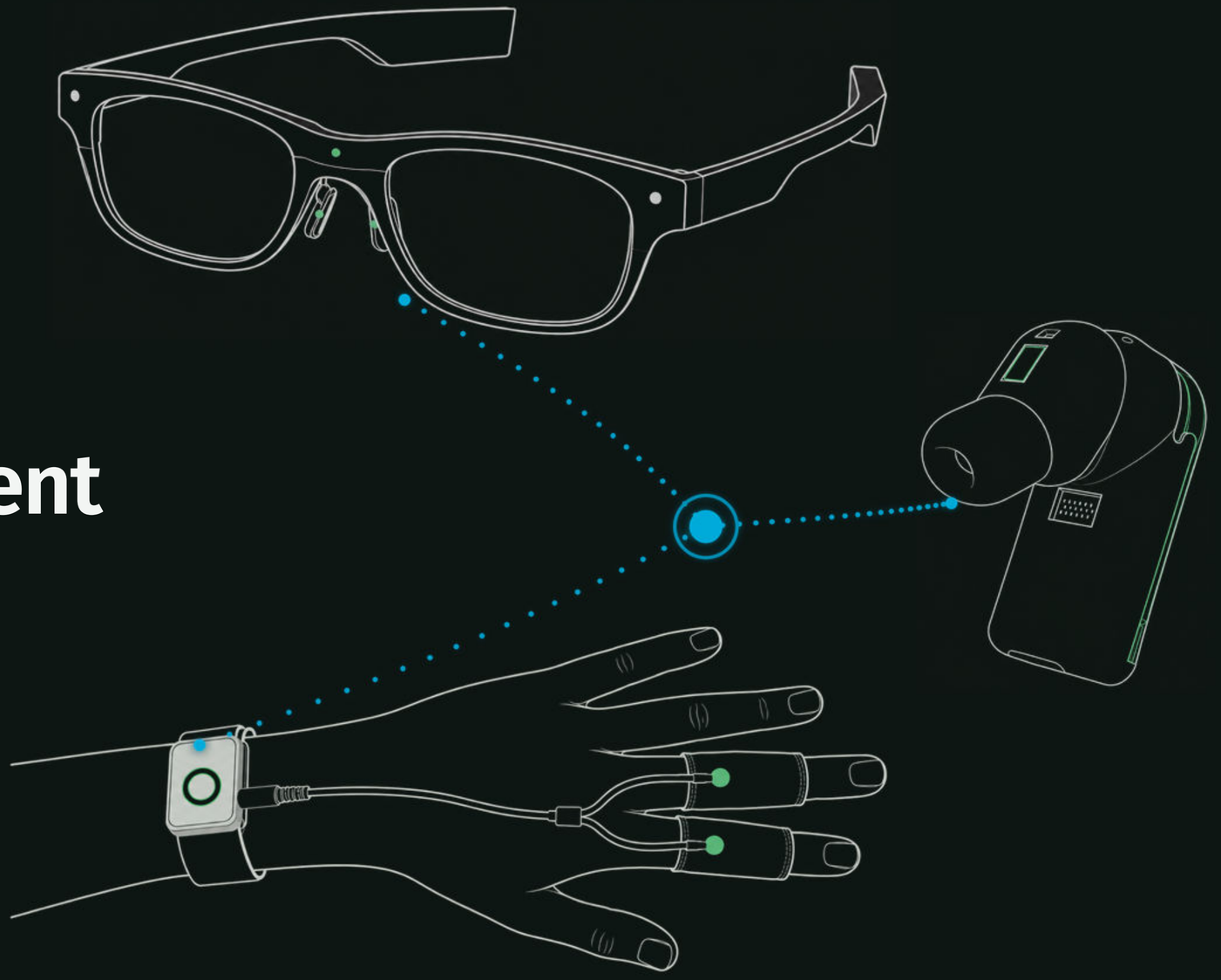






System Development

J!NS MEME and more





Is this how our brains work when we interact with the world?



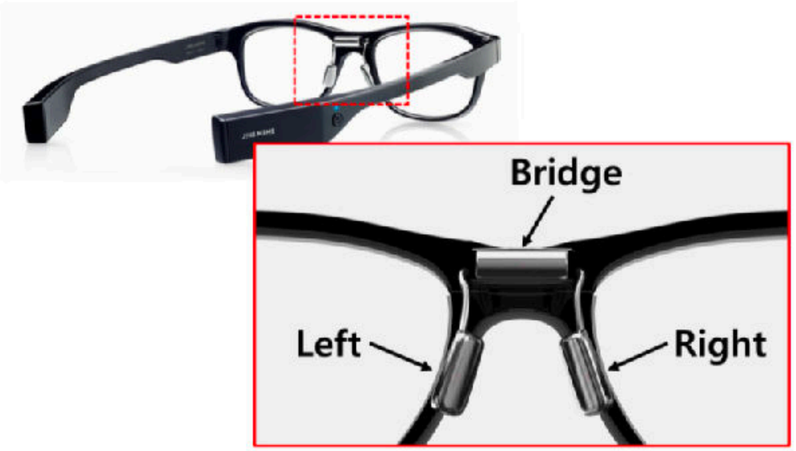
https://www.sciencedirect.com/science/article/pii/S1053811918306505

Cognitive Activity Tracking in Everyday Life



Glasses People Actually Wear All Day

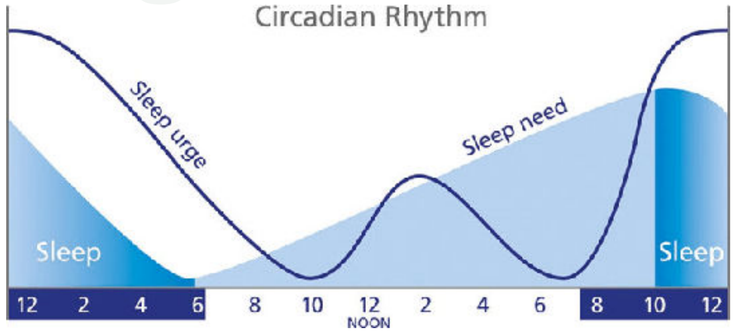


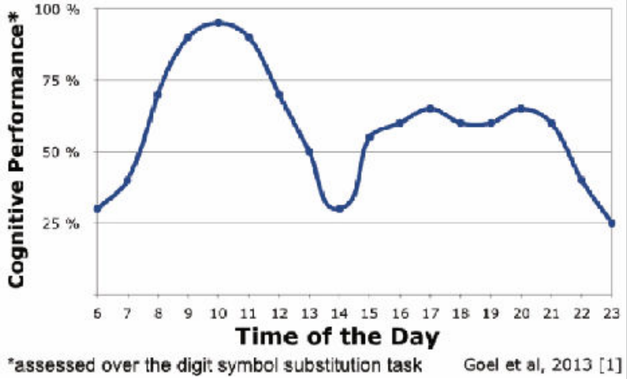


JINS MEME

3-axis accelerometer
3-axis gyroscope
2-axis EOG

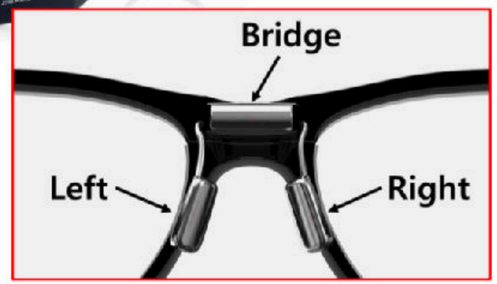
Cognitive Fluctuation Studies











Cognitive load, Circadian Rhythm, **Fatigue/Alterness**

20 participants, 3 weeks

At least 10 hours a day:

Electrooculography + Motion Sensors

“Groundtruth” every 2 hours:

Standard Cognitive Tests

Karolinska Sleepiness Scale

Psychomotor Vigilance Task (PVT)

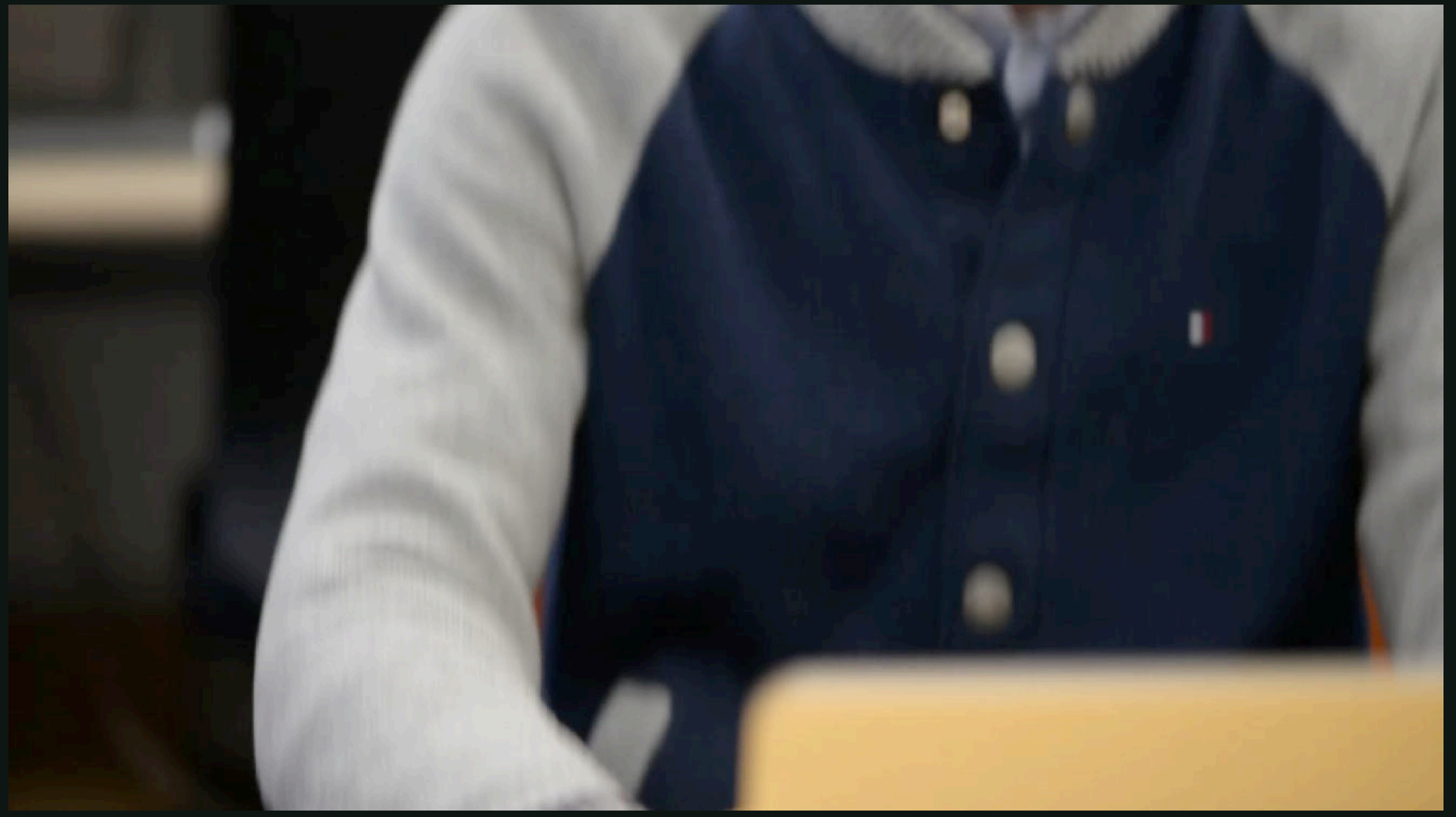
Stroop Test

Log Caffein intake and Naps

Sessions in the lab with FNIRS

Tag, Benjamin et all. "Continuous Alertness Assessments: Using EOG Glasses to Unobtrusively Monitor Fatigue Levels In-The-Wild." CHI, 2019.

Faces Without a Camera



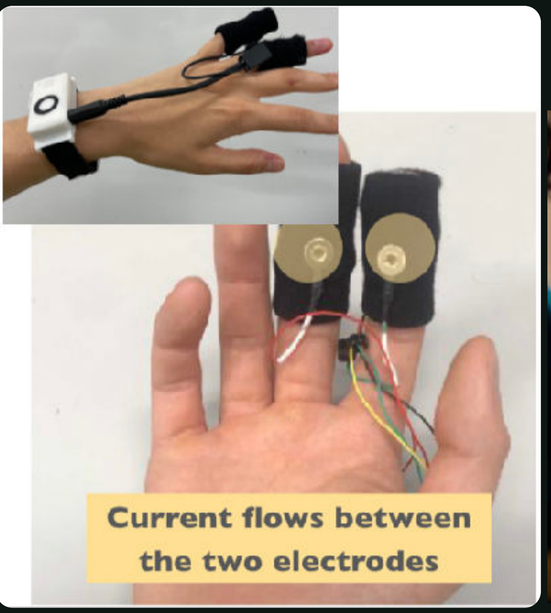
Evaluation of Facial Expression Recognition by A Smart Eyewear, ACM Transactions on Interactive Intelligent Systems, 2017.

Physiological Sensing Platform

Electrodermal Activity and Blood Volume Pulse

For larger scale deployments, live streaming

Electrodermal Activity and Blood Volume Pulse.







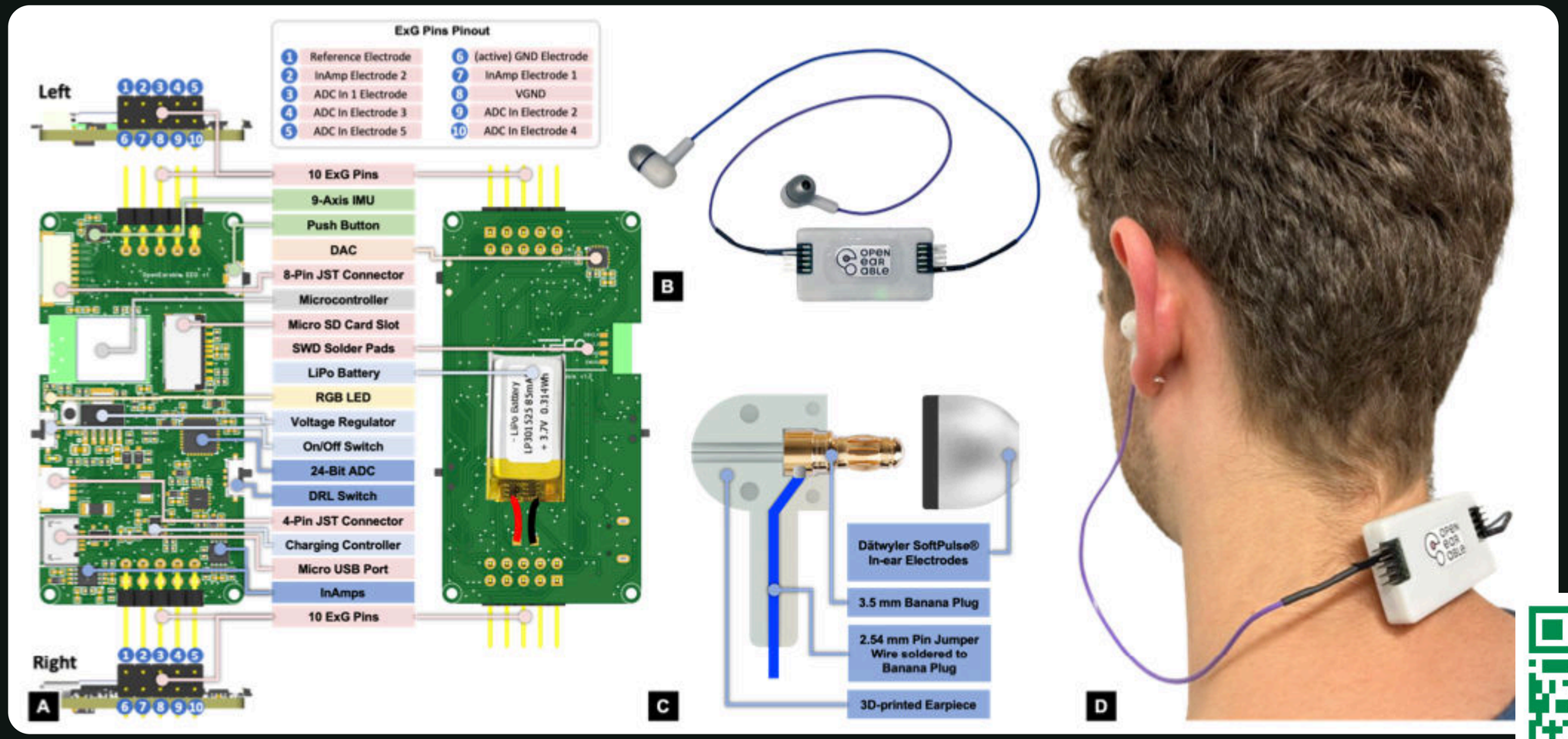


Activity | Systems | Augmented

Karlsruhe Institute of Technology

OpenEarable ExG

Open-Source Hardware for Ear-Based Biopotential Sensing Applications



Philipp Lepold, Tobias Röddiger, Tobias King, Kai Kunze, Christoph Maurer, and Michael Beigl. 2024. OpenEarable ExG: Open-Source Hardware for Ear-Based Biopotential Sensing Applications. In Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24). Association for Computing Machinery, New York, NY, USA, 916–920. https://doi.org/10.1145/3675094.3678480

Augmented

HCI Studies and beyond



From Interaction to Integration

Next Steps in Human-Computer Integration

Mainframe many : 1	→	**PC** 1 : 1	→	**Mobile** 1 : many

→ **Ubiquitous** many : many → **Integration** blurred boundary

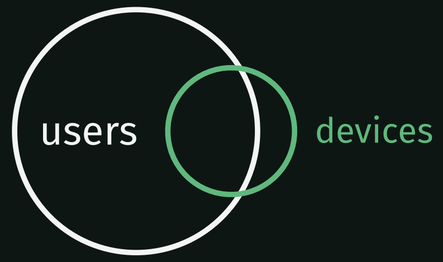
Users : machines per era



mobile



wearable



integrated

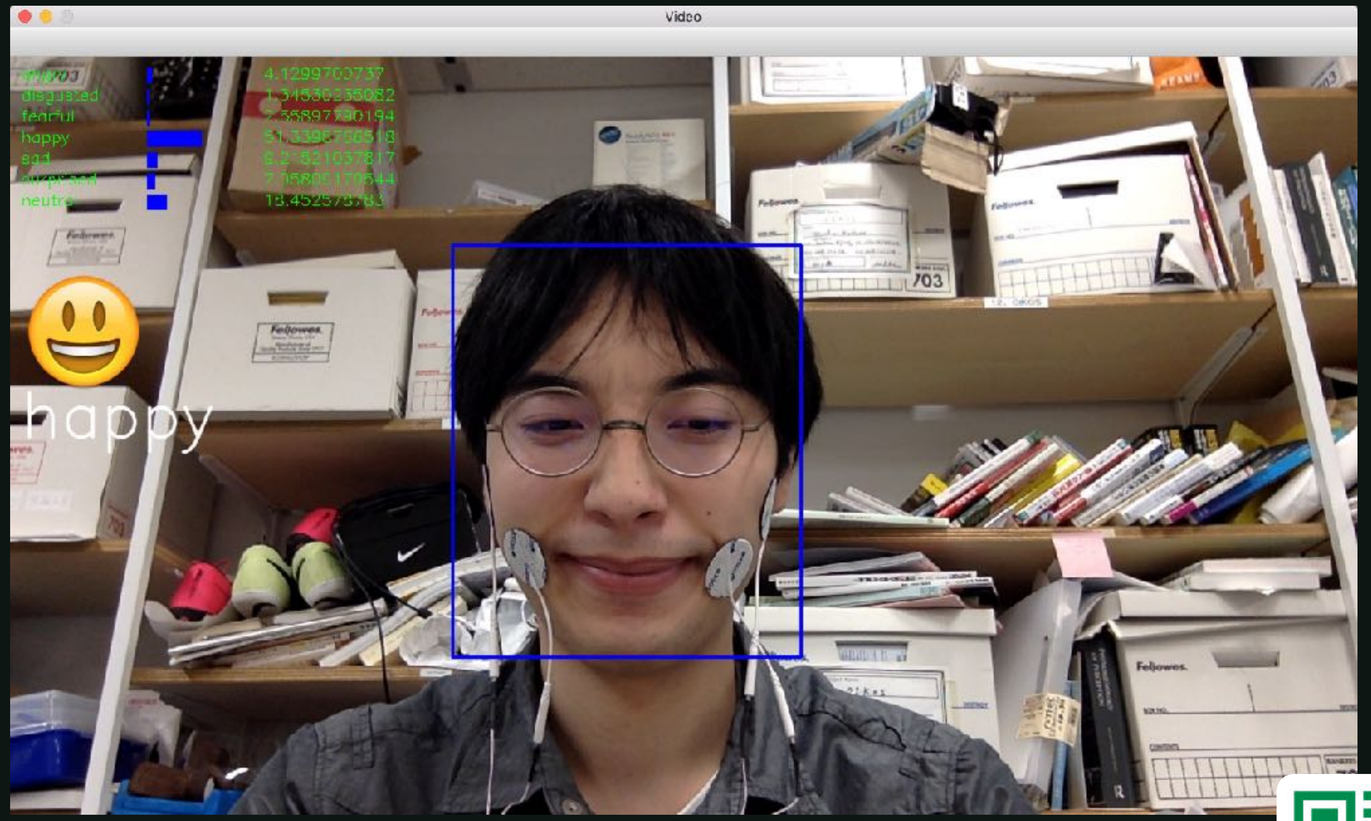
Mueller, Lopes, Strohmeier, ..., Kunze, ..., Maes, CHI 2020



When the Machine Moves Your Face

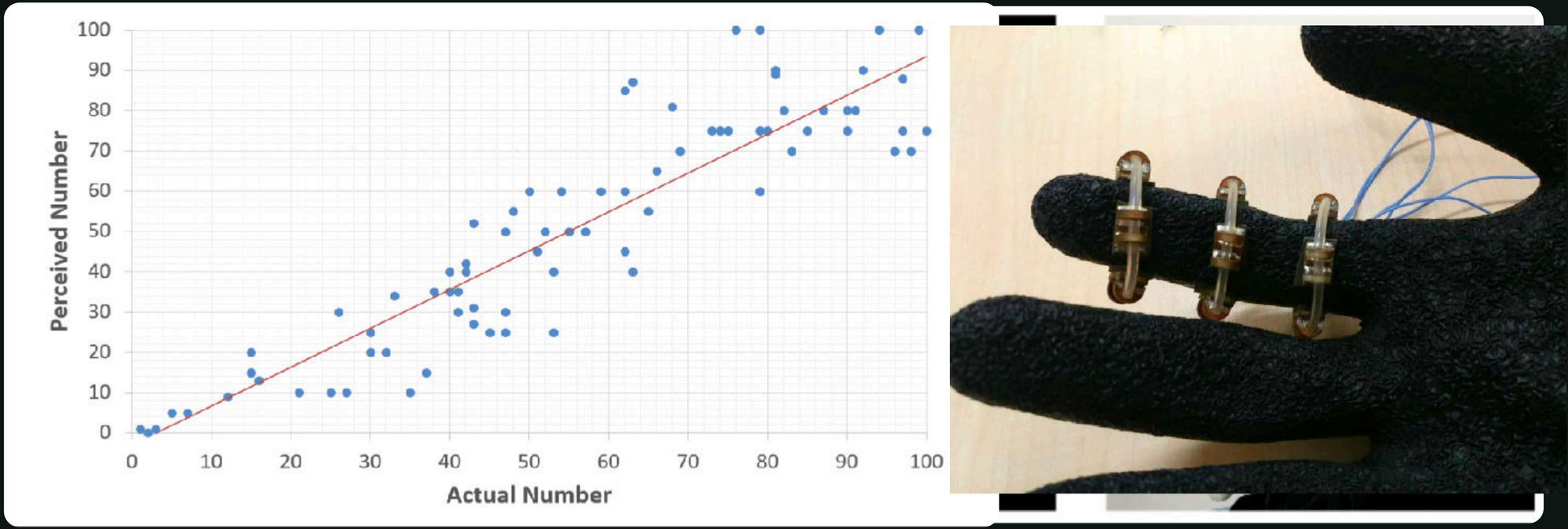
EMS smiles, recalibrated by a camera in a closed loop





Goto, Tag, Kunze, Dingler. Auto-calibrating EMS for emotional responses to media. Augmented Human 2018 · Lai et al. Eyewear to make me smile. TEI 2017

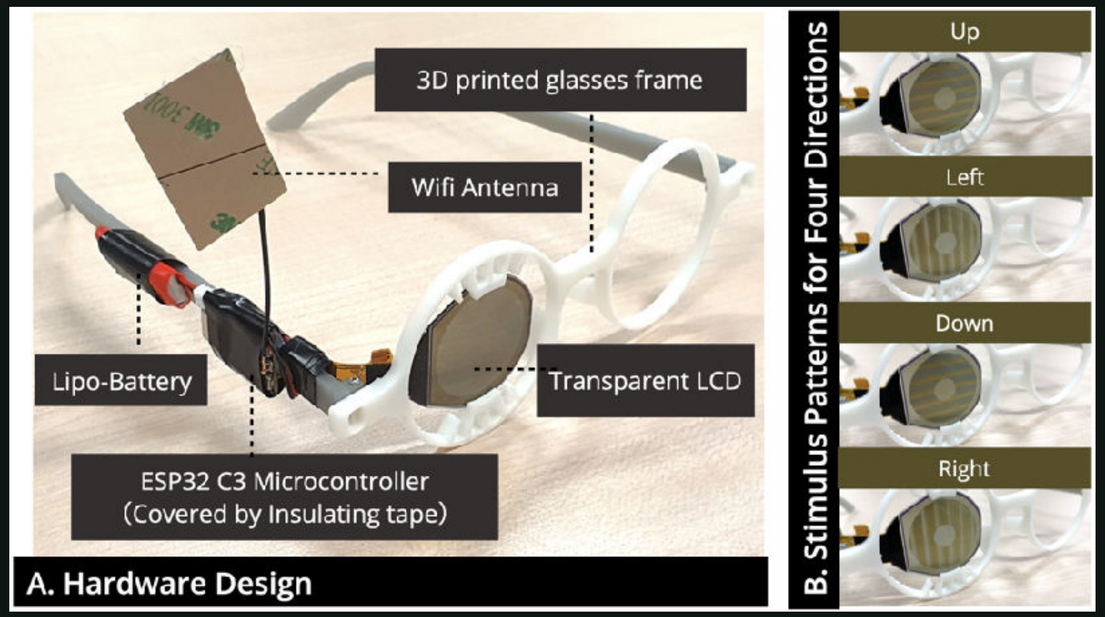
Feel a Progress Bar on Your Finger

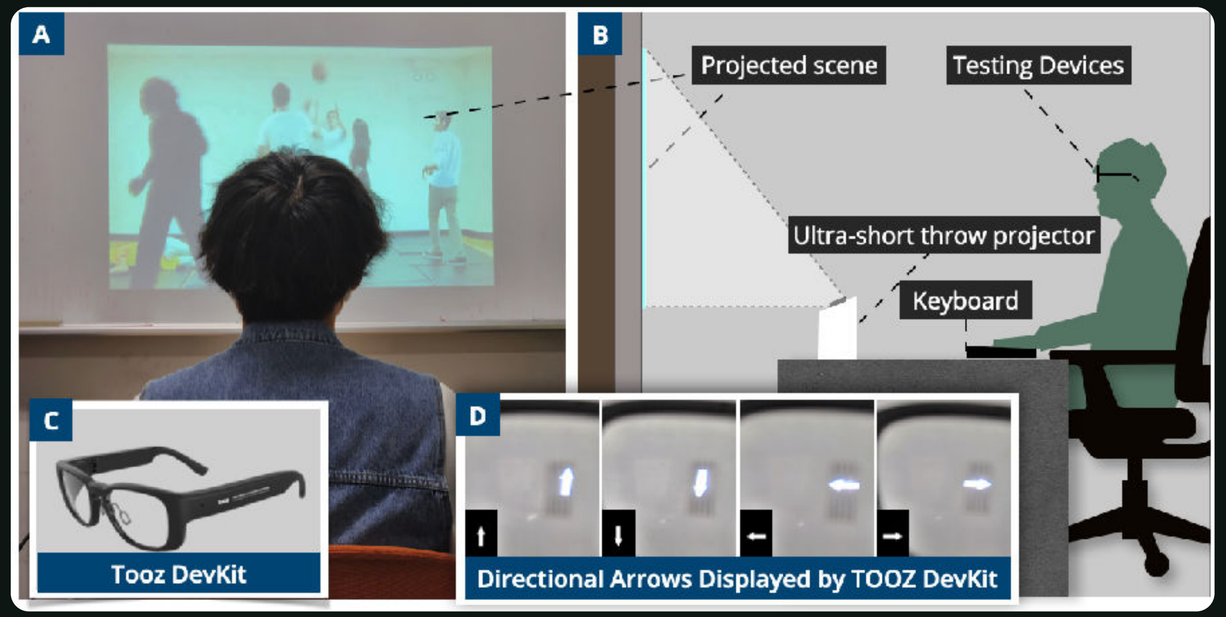


Chernyshov, George, et al. "Shape memory alloy wire actuators for soft, wearable haptic devices." Proceedings of the 2018 ACM International Symposium on Wearable Computers. ACM, 2018.

Directions Without Symbols

Beyond Symbols: Motion Cues in Peripheral Vision





Zhang, Chen, Huang, Huang, Starner, Kunze, Rekimoto, ISWC 2025

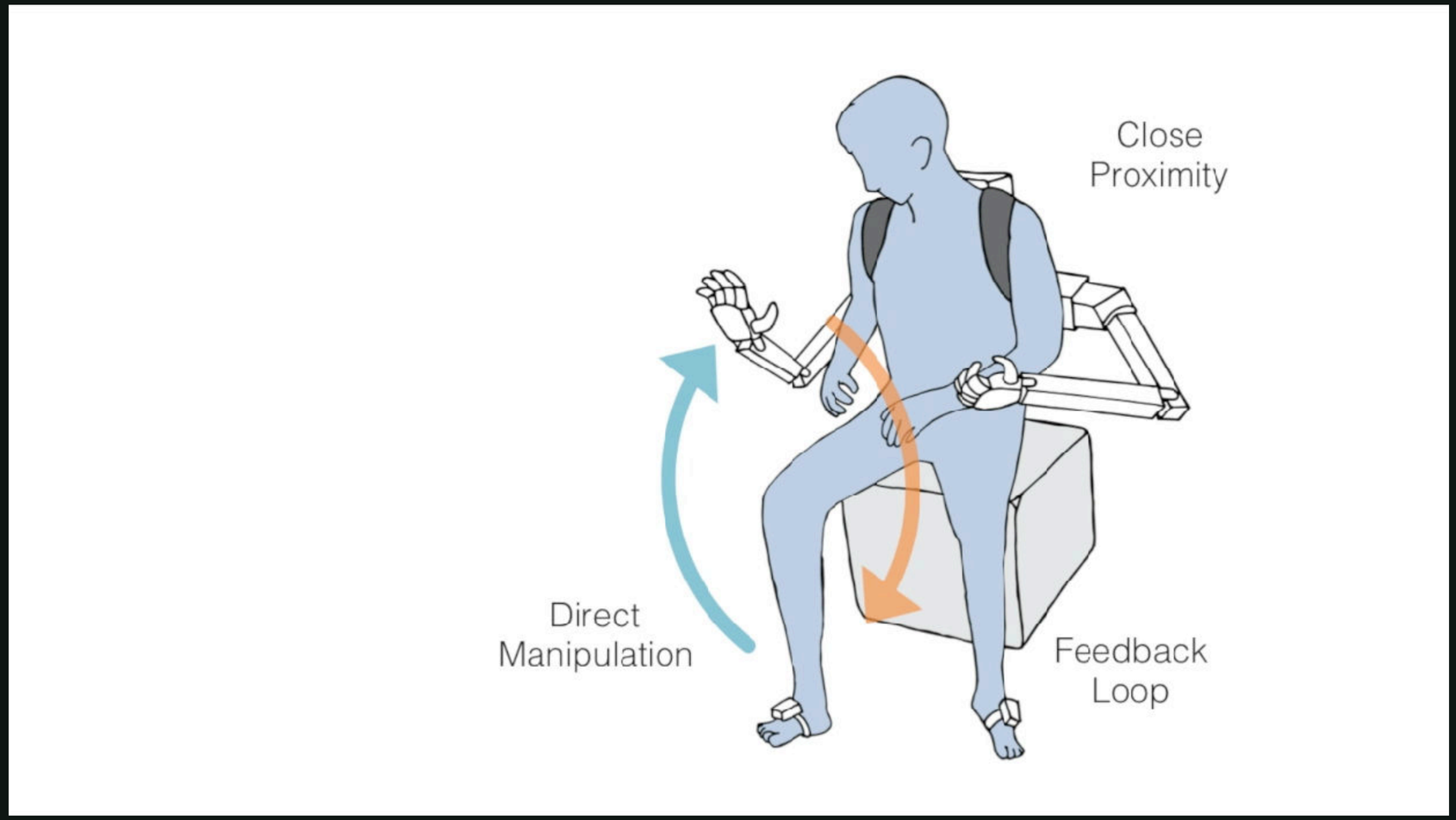
Directions Without Symbols

The Stimulus on the Monocle Display



Zhang, Chen, Huang, Huang, Starner, Kunze, Rekimoto, ISWC 2025

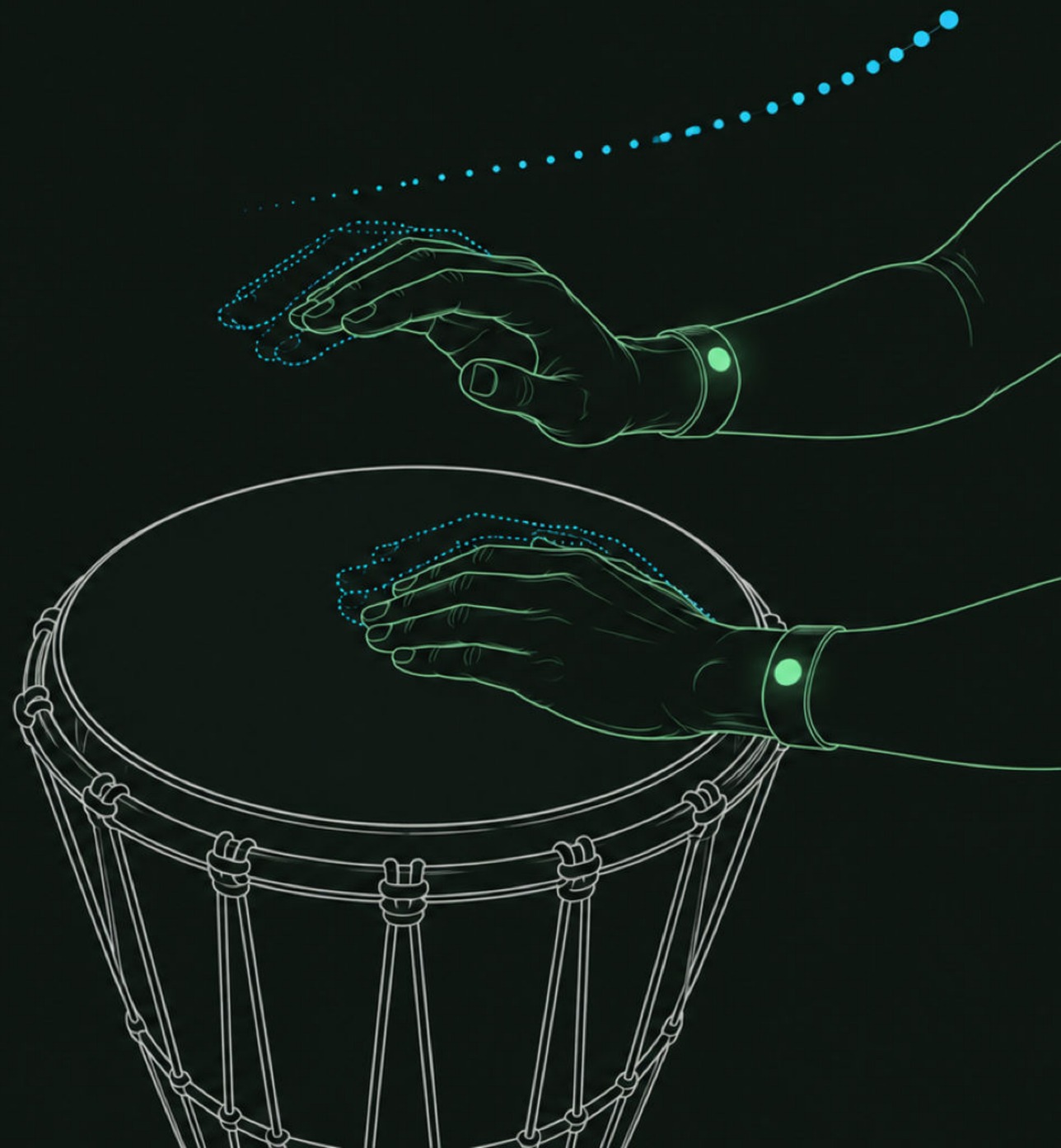
Explore how can Tech overlap with our bodies



Saraiji, M. H. D., et al. "MetaArmS: Body remapping using feet-controlled artificial arms." *The 31st Annual ACM Symposium on User Interface Software and Technology*. ACM, 2018.

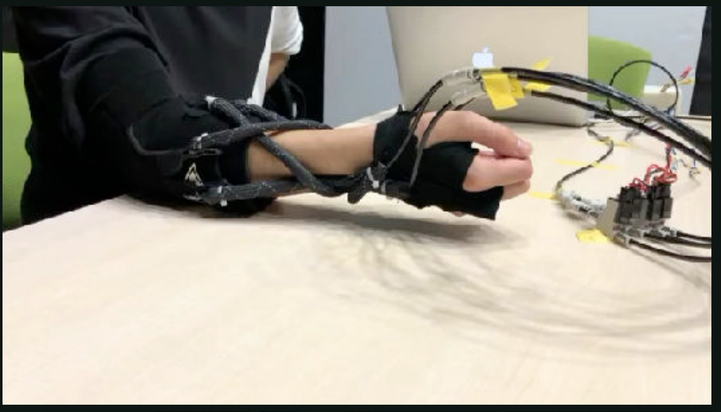
Skill

Which skills do I need, which should I develop?

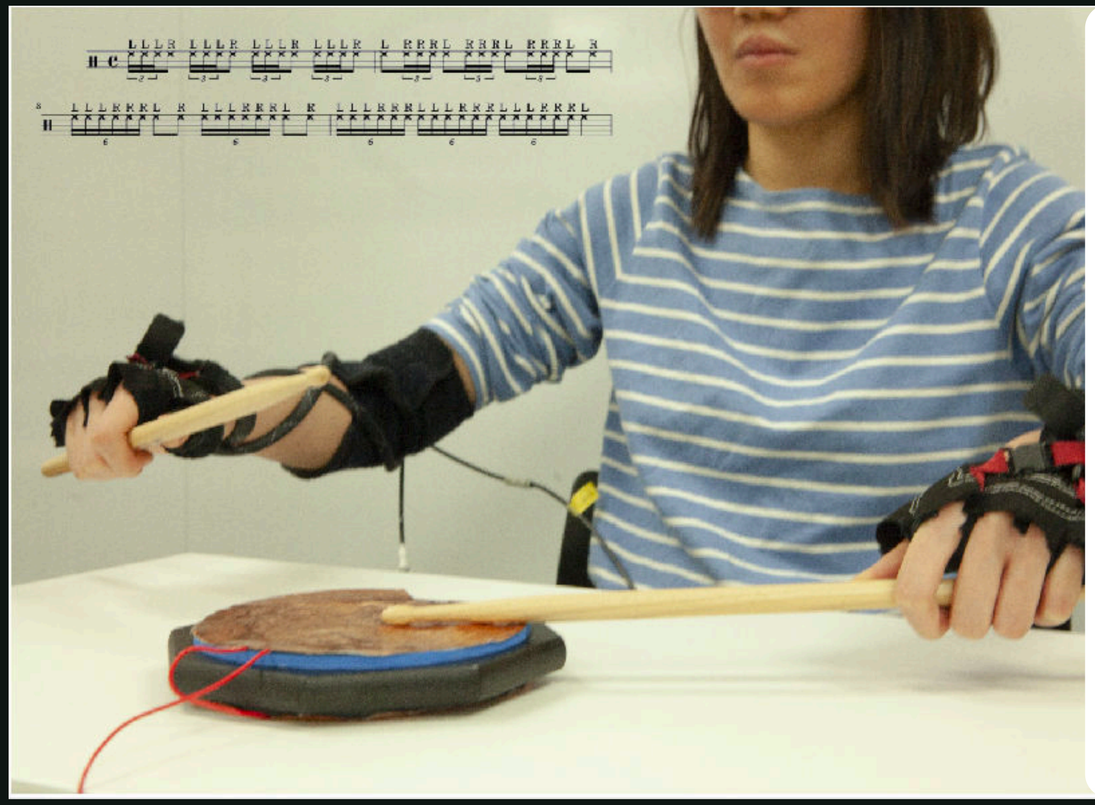


Improves Skill Acquisition in Percussions

Pneumatic Artificial Muscles







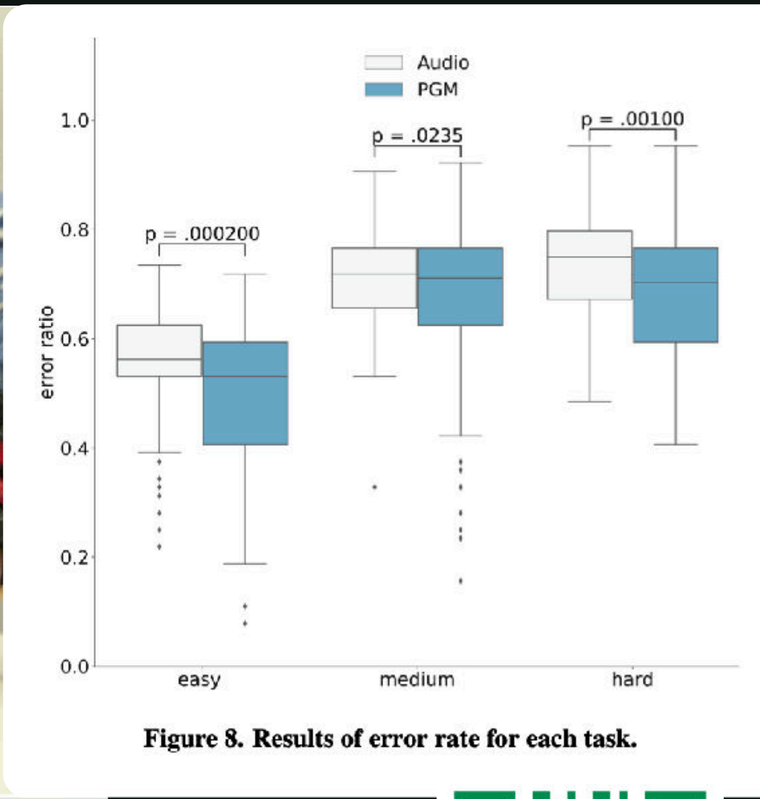


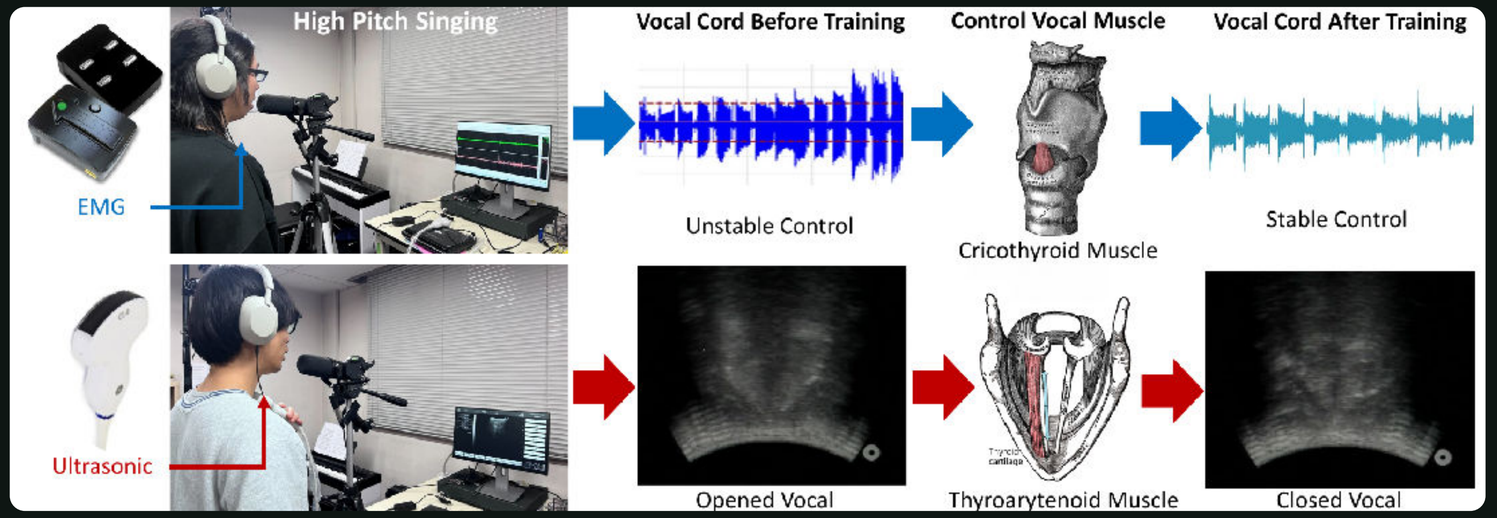
Figure 8. Results of error rate for each task.

Goto, Takashi, et al. "Accelerating Skill Acquisition of Two-Handed Drumming using Pneumatic Artificial Muscles." Proceedings of the Augmented Humans International Conference. 2020.

Sensing Your Vocals

EMG and ultrasound make the vocal cord muscles visible



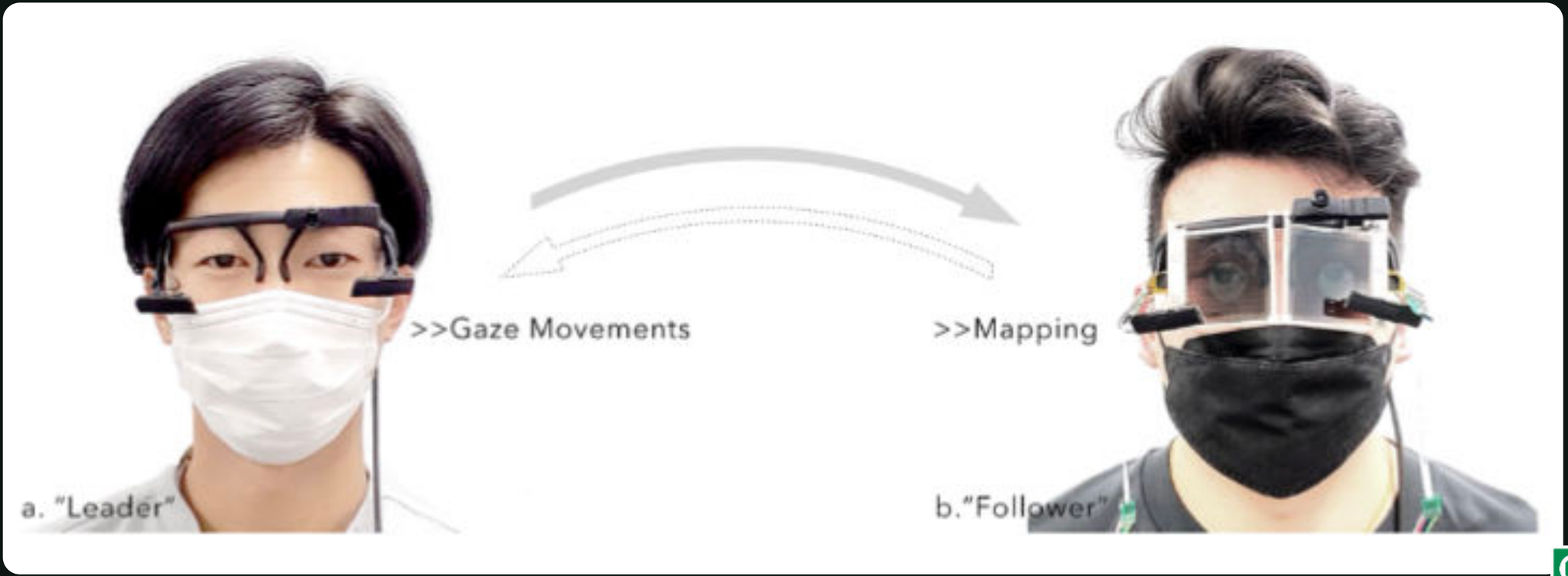


Honorable Mention · CHI 2026

Chen, Panskus, Wu, Peng, Saito, Kamiyama, Li, Liao, Marky, Kato, Koike, Kunze. Sensing Your Vocals: Exploring the Activity of Vocal Cord Muscles for Pitch Assessment Using Electromyography and Ultrasonography. CHI 2026. doi 10.1145/3772318.3790965

Can we share our visual perception with others?

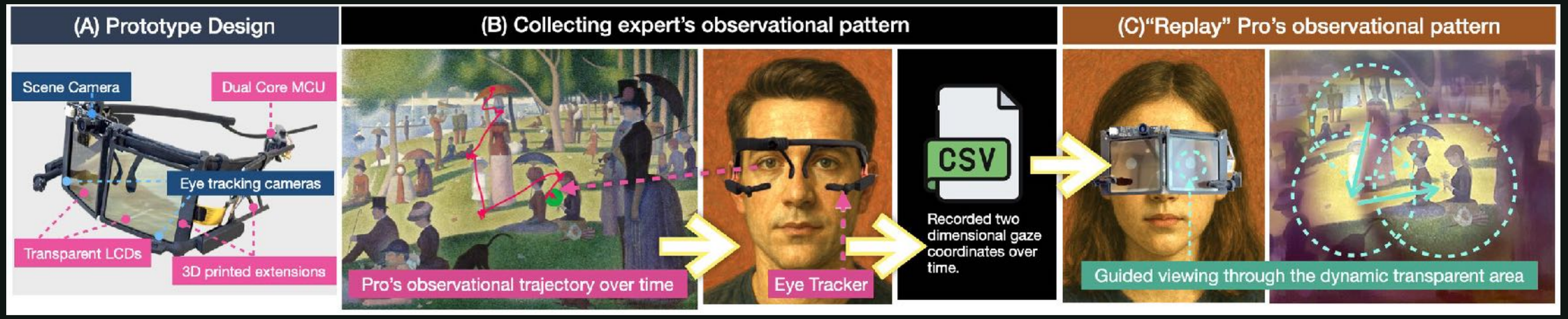
GazeSync (IUI 2022)



GazeSync: Eye Movement Transfer using an Optical Eye Tracker and Monochrome Liquid Crystal Displays. IUI 2022.

Learning How Professionals Observe

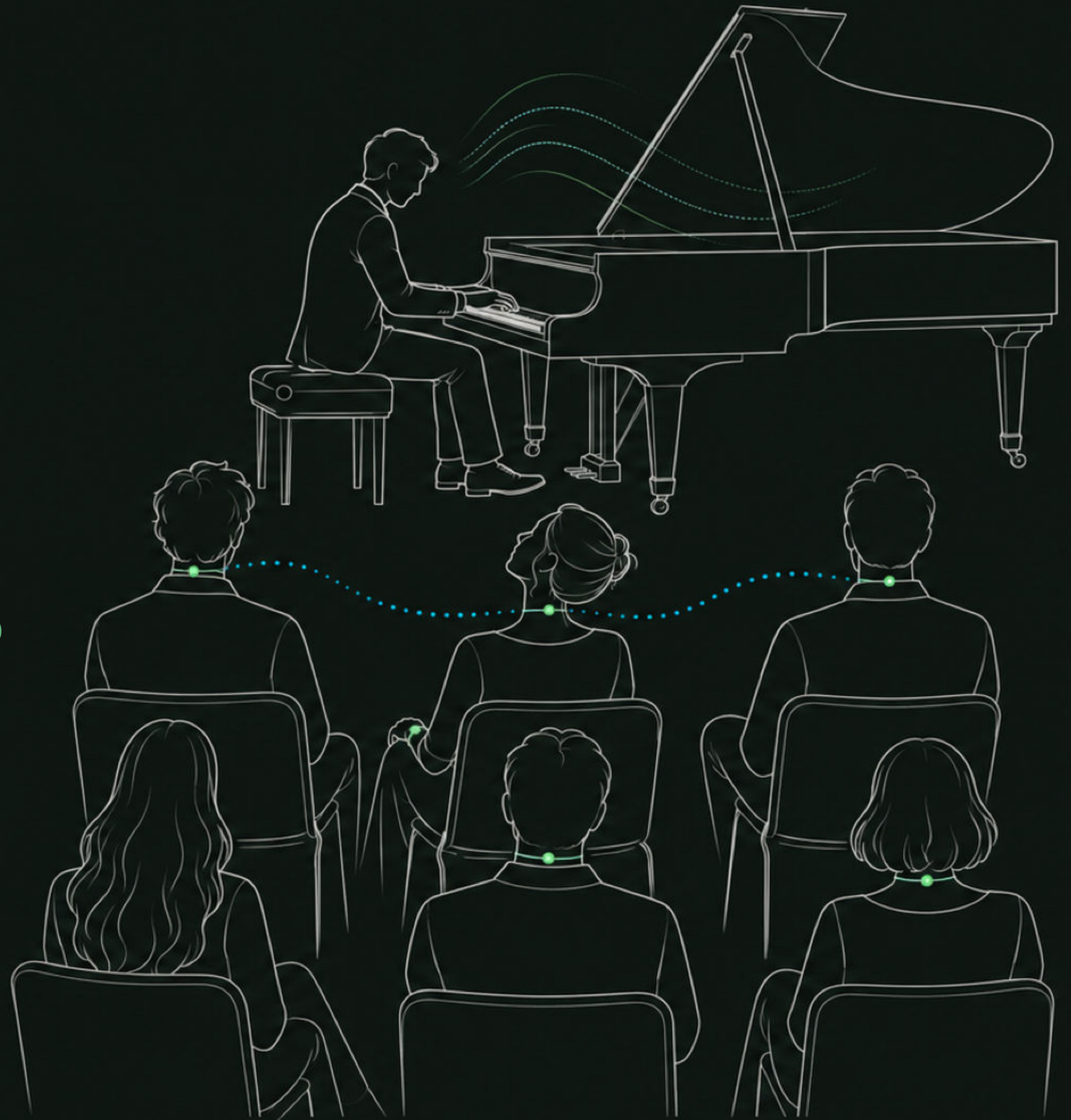
Pro's Eyes: follow an expert's gaze path



Zhang et al., SIGGRAPH Asia 2025 E-Tech

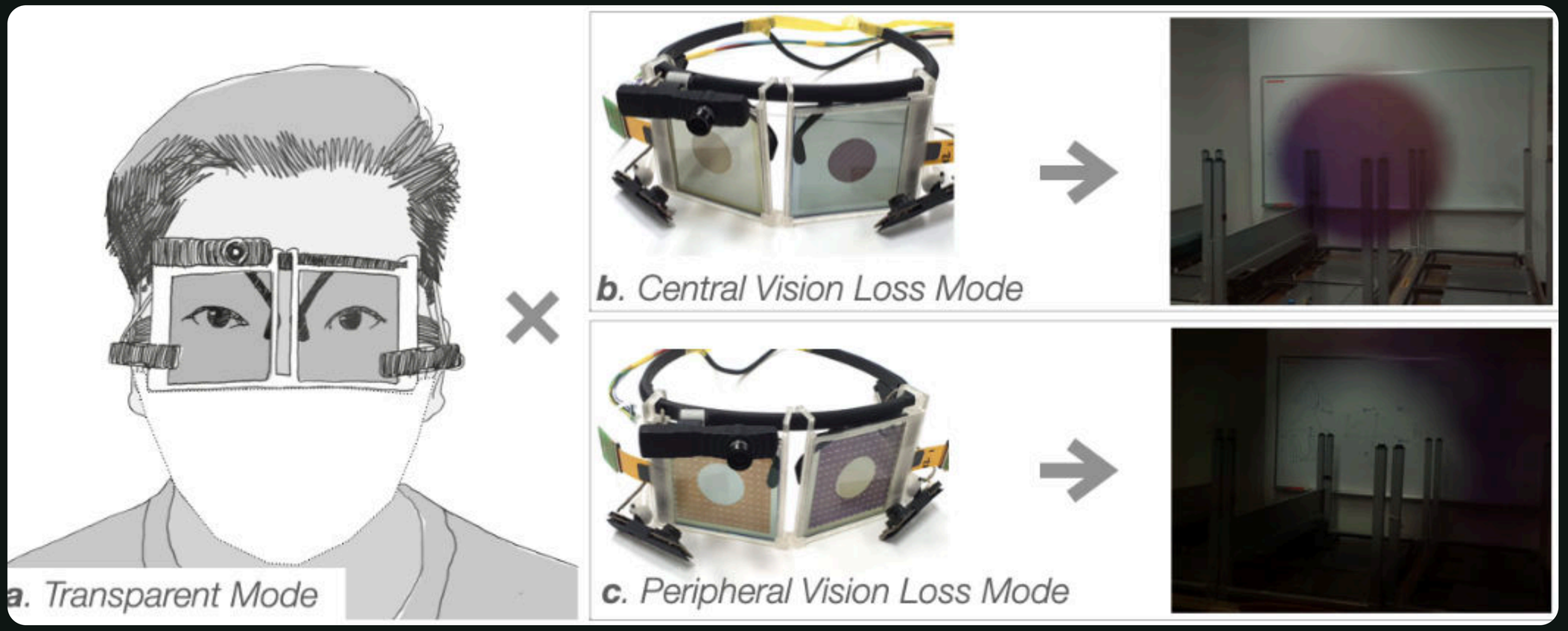
Experience

Can we design meaningful moments to share?



Visual Impairment Simulation

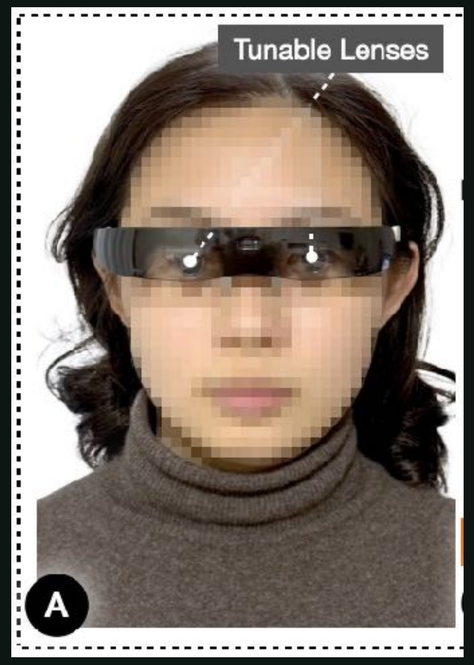
Seeing our Blind Spots: Smart Glasses-based Simulation to Increase Design Students' Awareness of Visual Impairment

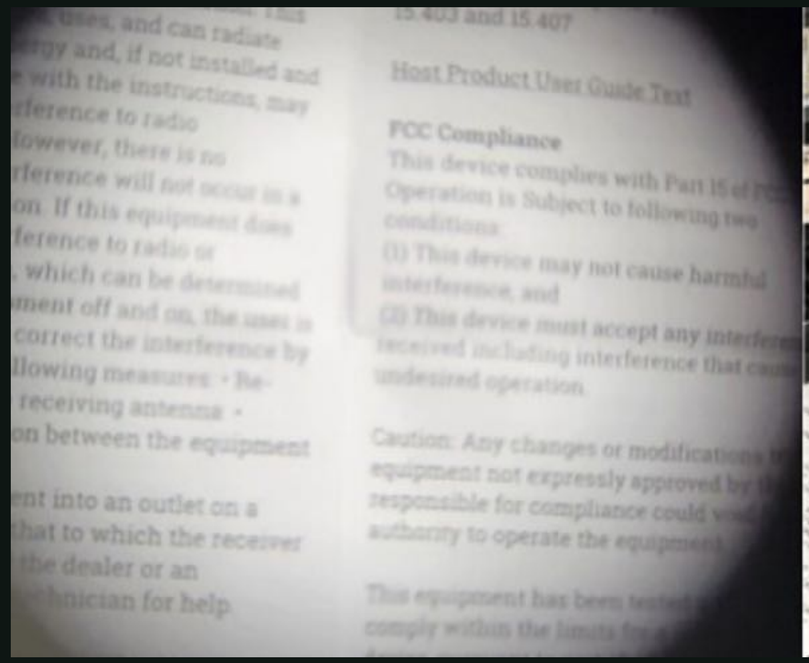


Zhang, Qing, et al. "Seeing our Blind Spots: Smart Glasses-based Simulation to Increase Design Students' Awareness of Visual Impairment." UIST 2022.

Simulating Ageing Eyes

OpticalAging: tunable lenses for age-inclusive design



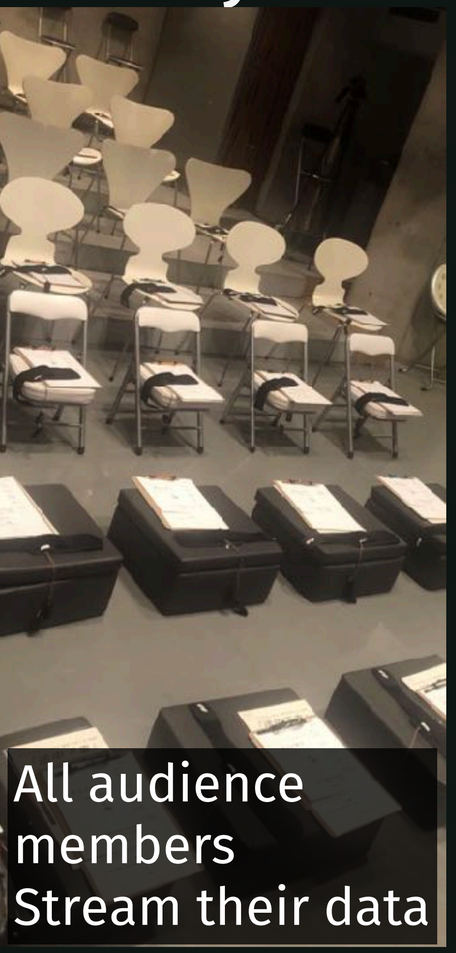




Zhang et al., Augmented Humans 2026

Putting the Audience on Stage

Why we chose the setup



All audience members Stream their data



Heart Rate/ Variability
Galvanic Skin Response



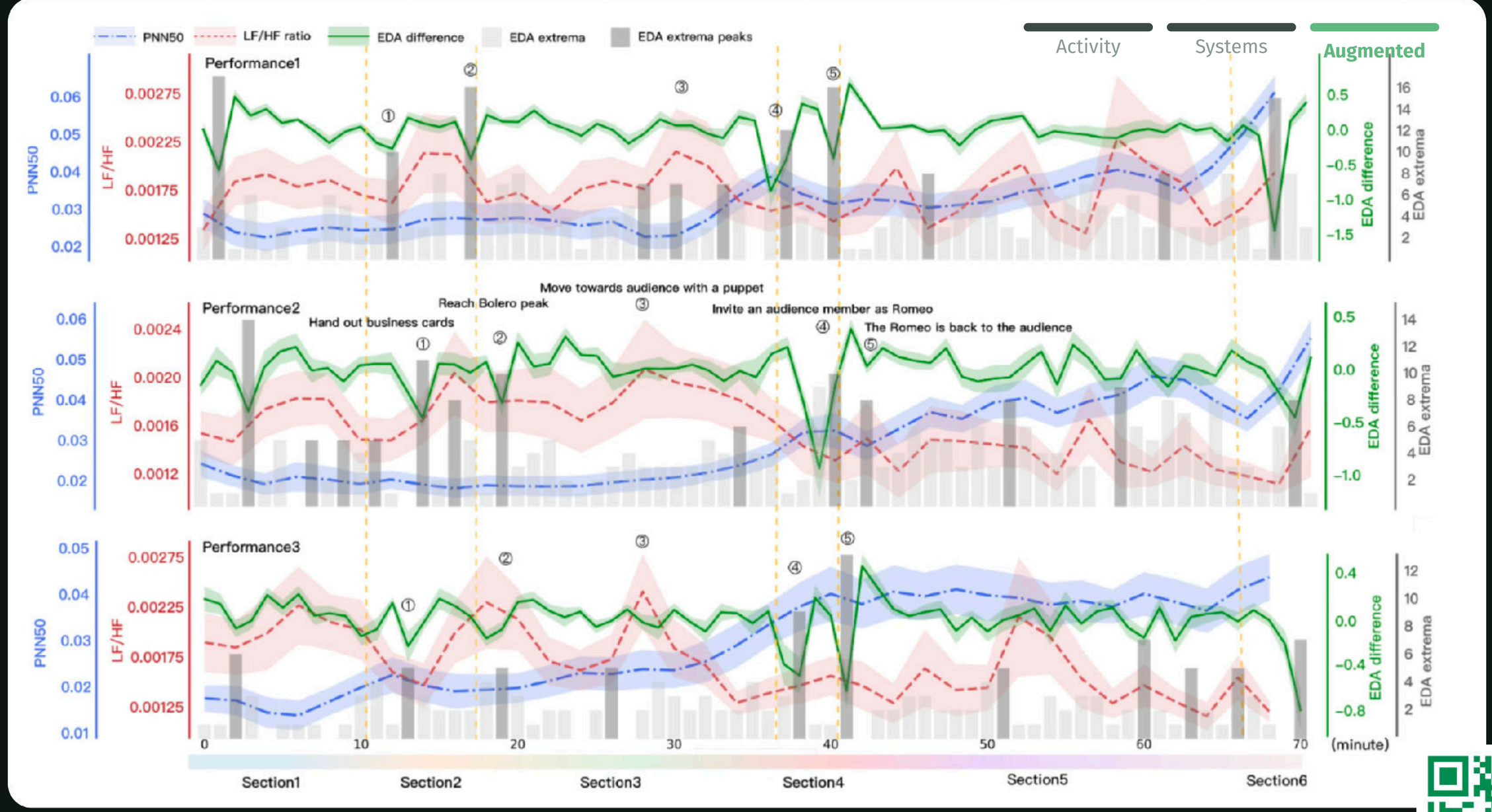
Projection, Lights, Sound change based on Sensor Data

Special thanks to: Dingding Zheng, George Chenyshov, Danny Hynds,

Linking Audience Physiology to Choreography.
TOCHI Journal 2023.
https://dl.acm.org/doi/10.1145/3557887

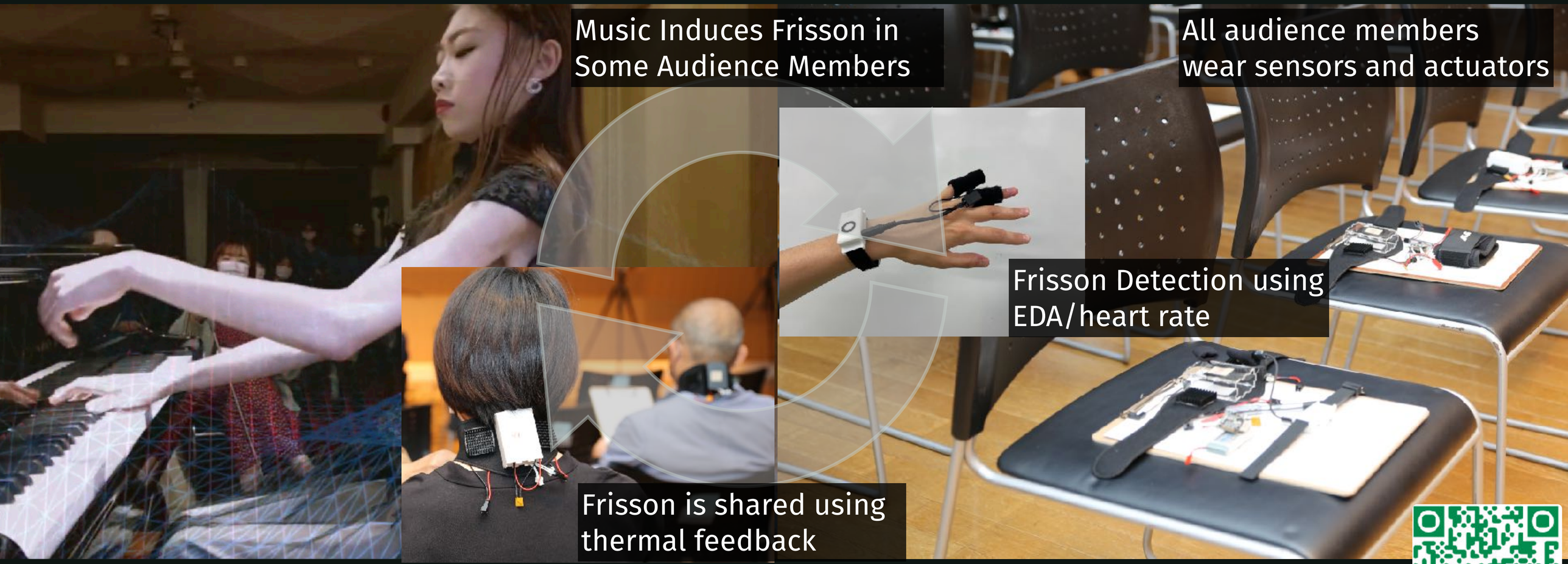


Our approach to the stage production relies on the transformation of the physical sensations of the dancers and the audience



Han, Chernyshov, Sugawa, Zheng, Hynds, Furukawa, Padovani Macieira, Marky, Minamizawa, Ward, Kunze. Linking Audience Physiology to Choreography. ACM TOCHI 30(1), 2023. doi 10.1145/3557887

Recording the Physiology in Classical Concerts



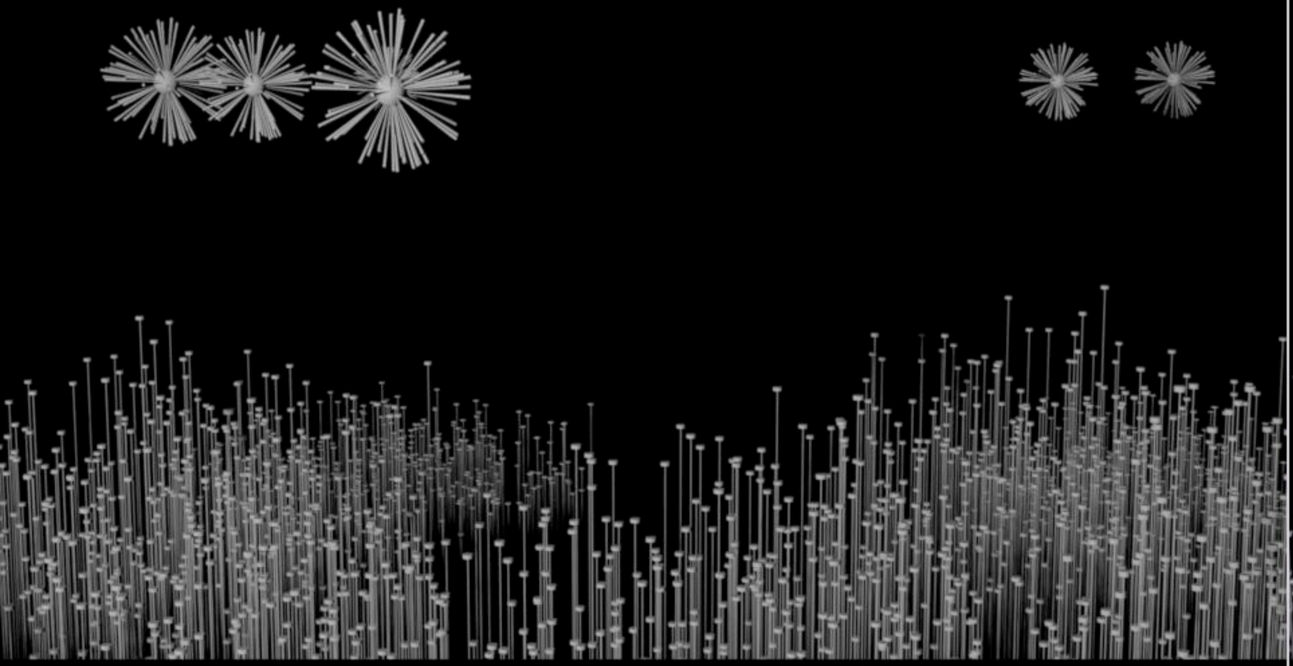
Yan He, George Chernyshov, Jiawen Han, Dingding Zheng, Ragnar Thomsen, Danny Hynds, Muyu Liu, Yuehui Yang, Yulan Ju, Yun Suen Pai, Kouta Minamizawa, Kai Kunze, and Jamie A. Ward. 2022. Frisson Waves: Exploring Automatic Detection, Triggering and Sharing of Aesthetic Chills in Music Performances. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 6, 3, Article 118 (September 2022), 23 pages. https://doi.org/10.1145/3550324

Physiological measurements to recreate collective experiences in VR



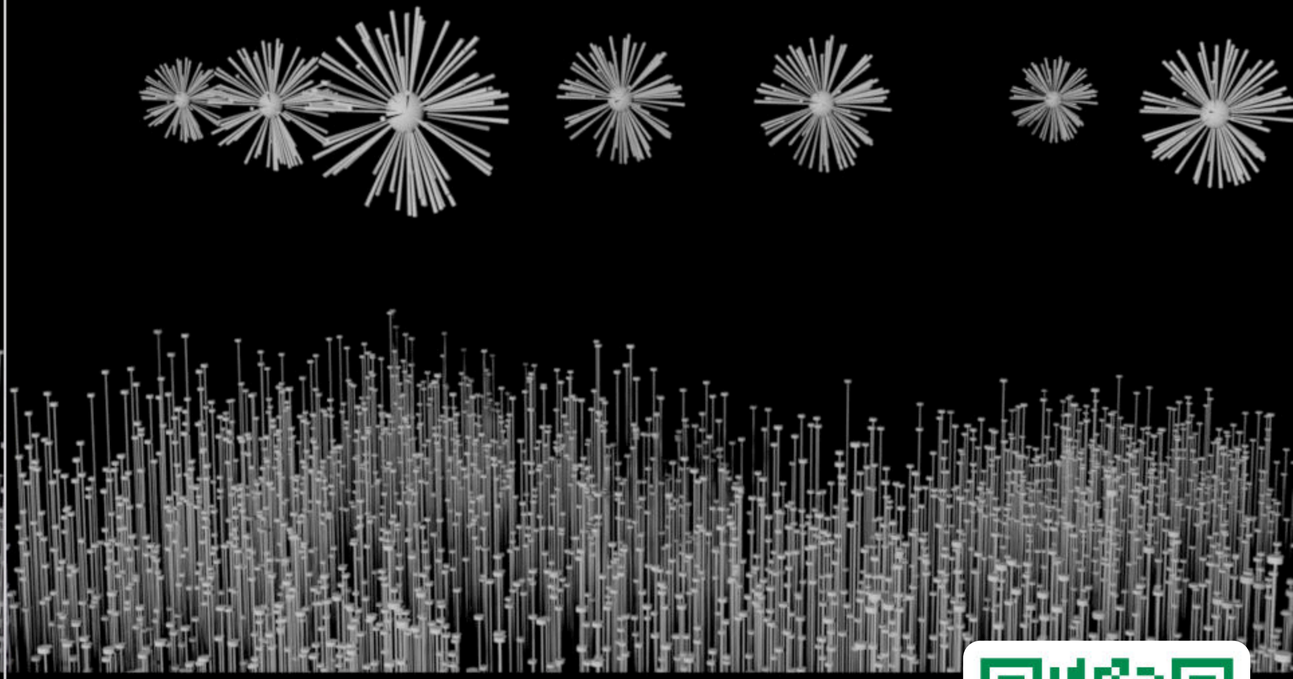
A Placebo Concert: The Placebo Effect for Visualization of Physiological Audience Data during Experience Recreation in Virtual Reality CHI 2025

Real Data



Visuals change dynamically based on
real audience data (EDA, BVP), reflecting authentic responses.

Fake Data

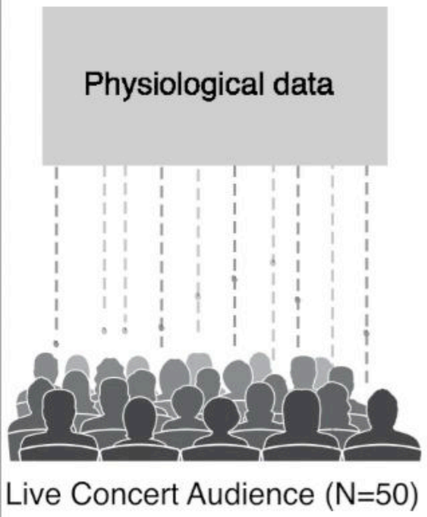


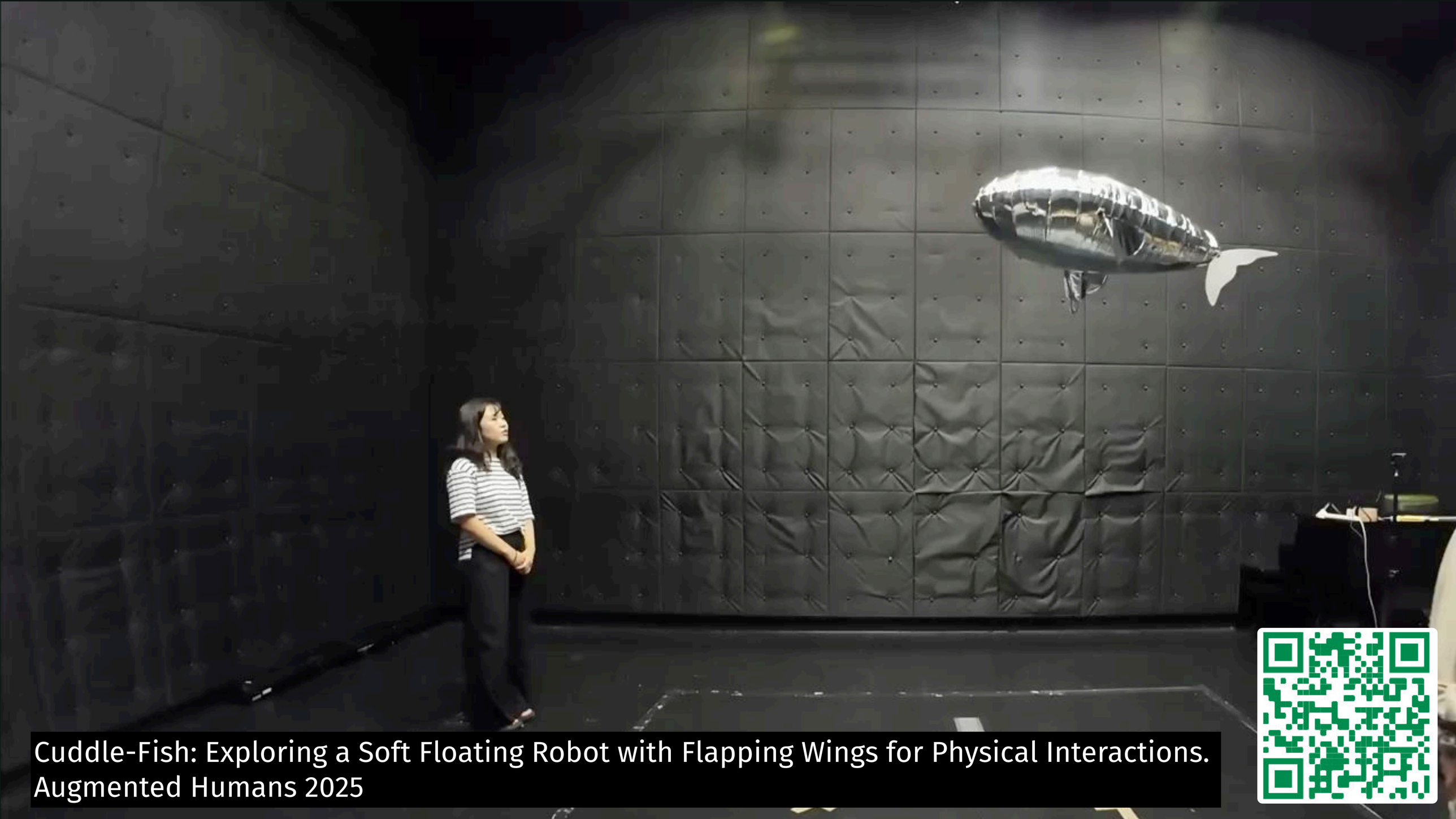
Visuals driven by random data,
showing ff

Meng et al., CHI 2025

User Study

Mixed-Design Study (n = 44)





Cuddle-Fish: Exploring a Soft Floating Robot with Flapping Wings for Physical Interactions.
Augmented Humans 2025

Floating Companion

Exploring Design Space for Soft Floating Robots in Indoor Environments


Mingyang Xu [1*]

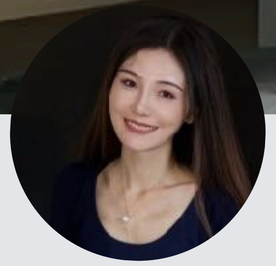

Yanheng Li [2*]


Burcu Nimet Dumlu [1]


RAY LC [2]


Giulia Barbareschi [3]


Matthias Hoppe [1]


Jie Li [4]

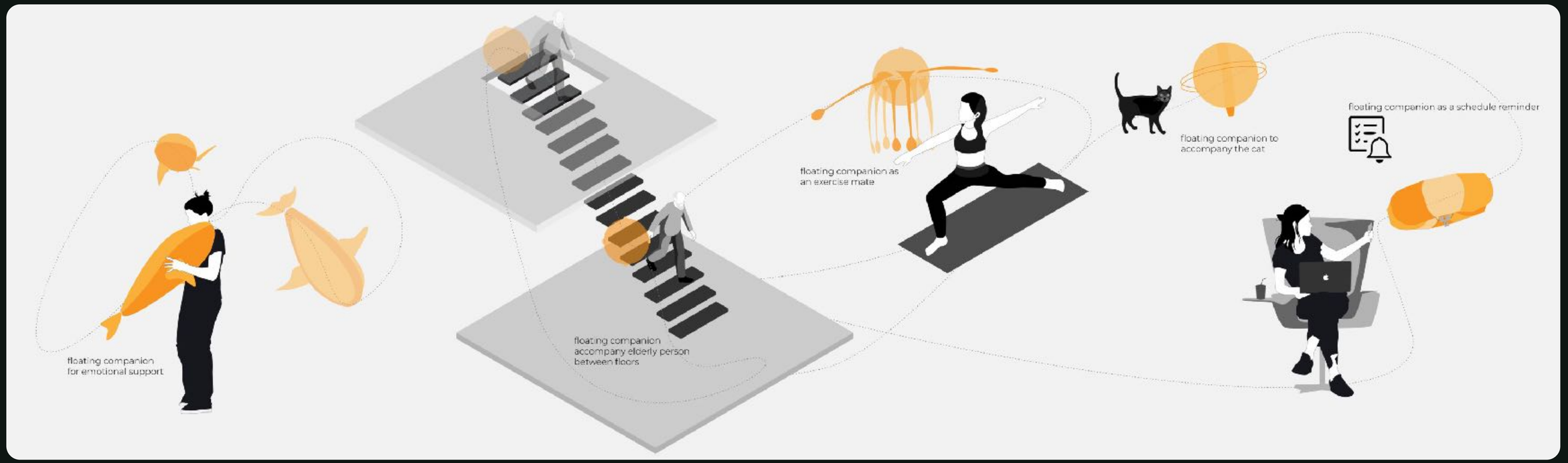

Kouta Minamizawa [1]


Kai Kunze [5]

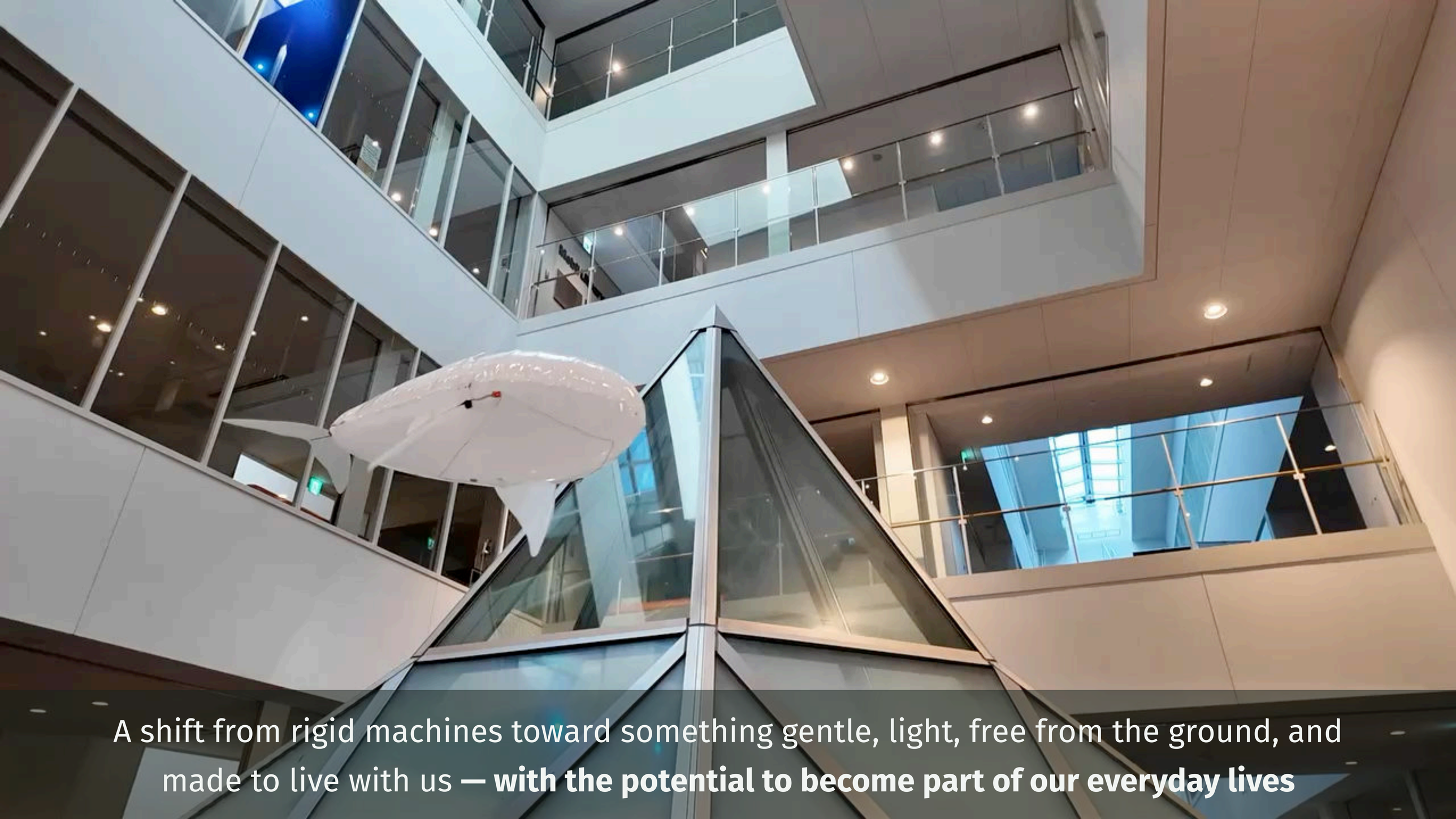
[1] Keio University Graduate School of Media Design
[2] City University of Hong Kong
[3] University of Duisburg-Essen
[4] Cake Researcher, MIT Media Lab
[5] TU Clausthal

Floating Companions in Everyday Life

Soft, lighter-than-air robots that share our rooms



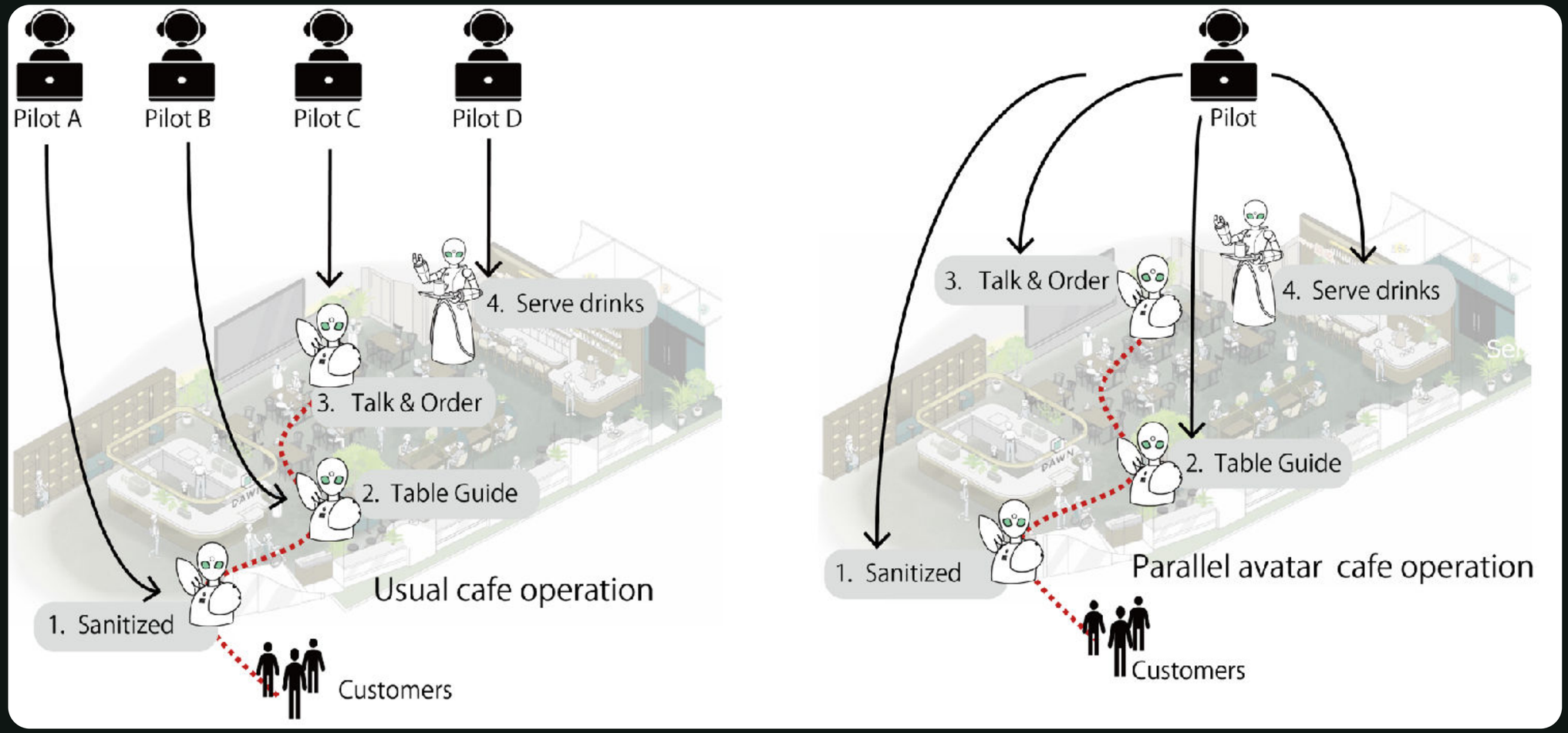
Xu, Li, Dumlu, LC, Barbareschi, Hoppe, Li, Minamizawa, Kunze. Floating Companion: Exploring Design Space for Soft Floating Robots in Indoor Environments. DIS 2026. doi 10.1145/3800645.3813051



A shift from rigid machines toward something gentle, light, free from the ground, and made to live with us **— with the potential to become part of our everyday lives**


DAWN
with TAILORED CAFE
スナック 織姫
Barista
06
Cafe
Lounge
30

Parallel Embodiment





Cybernetic being



“I am both here and there” Parallel Control of Multiple Robotic Avatars by Disabled Workers in a Café

Giulia Barbareschi, Midori Kawaguchi,Hiroki Kato, Masato Nagahiro, Kazuaki Takeuchi, Yoshifumi Shiiba, Shunichi Kasahara, Kai Kunze, Kouta Minamizawa



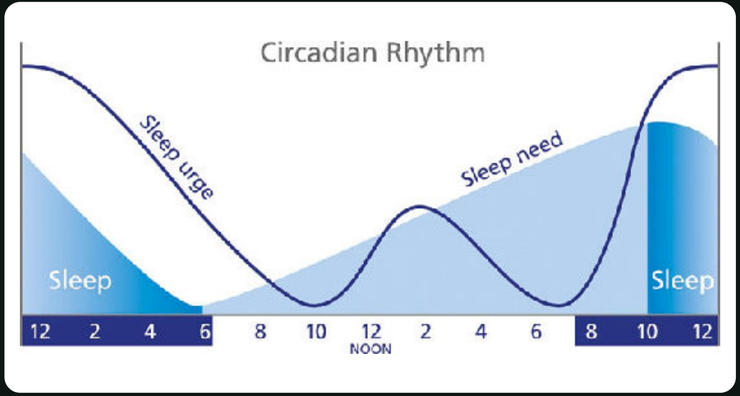






Design the Loop, Keep the Person

Three questions for augmentation with AI







Agency

Does it sense how you engage, and still feel like you doing it?

The system senses and suggests. You see it first, and you steer.

Skill

Which skills do you need? Which should you develop?

Take the device off. What is left should be yours.

Experience

Can we design meaningful moments with others?

Share what people feel, and share it honestly.

Tag et al., CHI 2019 · Goto et al., Augmented Humans 2020 · He et al., IMWUT 2022 · Meng et al., CHI 2025 · Bainbridge, Ironies of automation, 1983

What do you still want to ...

learn?

understand?

experience?

Thanks

To the people who actually did the work



Mingyang Xu



Danyang Peng



Kanyu Chen



Keio University · Graduate School of Media Design · GEIST



TU Clausthal · Human-Centered Information Systems · GEIST

Comments, remarks, violent dissent?

https://geist.pro